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Speculators, commodities and cross-market linkages☆

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ABSTRACT

We use a unique, non-public dataset of trader positions in 17 U.S. commodity futures markets to provide novel evidence on those markets' financialization in the past decade. We then show that the correlation between the rates of return on investible commodity and equity indices rises amid greater participation by speculators generally, hedge funds especially, and hedge funds that hold positions in both equity and commodity futures markets in particular. We find no such relationship for commodity swap dealers, including index traders (CITs). The predictive power of hedge fund positions is weaker in periods of generalized financial market stress. Our results support the notion that *who* trades helps

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Commodity index traders
Dynamic conditional correlations (DCCs)

predict the joint distribution of commodity and equity returns. We find qualitatively similar but statistically weaker results using a proxy for hedge fund activity based on publicly available data.

1. Introduction

In the past decade, financial institutions such as hedge funds and commodity index funds have assumed an ever greater role in commodity futures markets. We use a unique, non-public dataset from the U.S. Commodity Futures Trading Commission (CFTC) to provide novel evidence of this "financialization" and to document its association with an important aspect of the joint distribution of commodity and equity returns.

Since Friedman (1953), a large body of work has investigated whether the composition of trading activity (i.e., *who* trades) matters for asset pricing. On the one hand, many traders face constraints on their choices of trading strategies. Hence, the arrival of traders facing fewer restrictions should in theory help alleviate price discrepancies (Rahi and Zigrand, 2009) and improve risk transfers across markets (Başak and Croitoru, 2006). Insofar as hedge funds are less constrained than other investors (Teo, 2009) and as commodity markets have historically been partly segmented from other financial markets (Bessembinder, 1992), this theoretical argument suggests that increased hedge fund activity in commodity markets could strengthen cross-market linkages – especially if this increase reflects the arrival of traders not previously active in those markets.

On the other hand, the same financial traders who may help link different asset markets in normal times often face, in periods of financial market stress, borrowing constraints and sundry other pressures to liquidate risky positions. If so, then their exit from "satellite" markets (such as emerging markets or commodity markets) during a period of stress in a "central" market (such as the U.S. equity market) could in theory bring about cross-market contagion – see, e.g., Kyle and Xiong (2001), Kodres and Pritsker (2002), Broner et al. (2006), Pavlova and Rigobon (2008), and Danielsson et al. (2011, 2013). Then again, in the aftermath of the initial shock, reduced activity by value arbitrageurs or convergence traders could in turn lead to a decoupling of markets that they had helped link in the first place.

In this paper, we present empirical evidence that hedge fund positions in commodity futures markets help predict the strength of commodity–equity cross-market linkages. Controlling for market fundamentals, we document that commodity–equity return co-movements are positively related to the extent of commodity market participation by financial speculators as a whole and hedge funds especially – notably by hedge funds that hold positions in *both* equity and commodity futures markets. We find no such relationship for a measure of commodity index trading activity. The predictive power of hedge fund activity varies over time. Strikingly, it is weaker during periods of financial market turmoil. Our findings contribute to the debate on the implications of the "financialization" of commodity markets.

A major innovation of our paper is its dataset. In general, testing whether the activities of specific types of traders help predict cross-market linkages is difficult because doing so requires detailed information about the trading activities of all market participants as well as knowledge of each participant's main motive for trading. We overcome this empirical pitfall by constructing a daily dataset of individual trader positions in 17 U.S. commodity and equity futures markets between July 2000 and March 2010. The underlying raw data, which are non-public, originate from the CFTC's large trader reporting system (LTRS). The LTRS contains information on the end-of-day positions of every large trader in each of these 17 markets and, crucially, information on each trader's main line of business and purpose for trading. The LTRS information covers 70–90% of the open interest in the main U.S. commodity futures markets (Fishe and Smith, 2012).

We make three contributions to the debate on the financialization of commodities. First, we provide a decade's worth of novel data on the growing importance of different types of financial traders in a large number of U.S. commodity-futures markets. Second, we provide the first evidence about the extent to which different kinds of traders in those markets (in particular, hedge funds) also trade equity futures. We document that such cross-market trading has grown substantially in the past decade. Third, we use this heretofore unavailable information to shed light on the relationship between financialization and cross-market linkages.

Specifically, we show that variations in the make-up of the commodity futures open interest help predict long-term fluctuations in commodity–equity return co-movements. We employ ARDL regressions, using lagged values of the variables in the regression to tackle serial autocorrelation and possible endogeneity issues (which arise from the possibility that speculative activity could result from high volatility and correlations, rather than the other way around). We find that a 1% increase in hedge funds' share of the commodity futures open interest is associated, *ceteris paribus*, with an increase in equity–commodity return correlations of about 4%.

In contrast, we find that the positions of commodity swap dealers and index traders hold little predictive power for commodity–equity dynamic conditional correlations. Indeed, it is not just changes in the *overall* amount of financial activity in commodity futures markets that helps predict the observed correlation patterns. Instead, we trace that predictive power to the positions of hedge funds and, especially (and quite intuitively), of the subset of hedge funds that are active in *both* equity and commodity futures markets.

Turning to the impact of financial turmoil on cross-market linkages, we identify two patterns. First, we show that equity–commodity co-movements are positively related to the TED spread (our proxy for financial-market stress). From July 2000 until Lehman Brothers' demise in 2008, a 1% TED spread increase was associated with an approximately 0.20% increase in the dynamic equity–commodity correlation estimate. Intuitively, hedge funds could be an important transmission channel of negative equity market shocks into the commodity space. In fact, the sign of an interaction term we use to capture the behavior of hedge funds during financial stress ("high TED") episodes is statistically significant and negative. In other words, the predictive power of hedge fund activity is reduced during periods of global market stress.

Second, we document that commodity–equity correlations soared after the demise of Lehman Brothers and remained exceptionally high through the end of our sample in 2010. Over and above the explanatory powers of the TED spread and hedge fund positions, a time dummy capturing the post-Lehman period (September 2008–March 2010) is highly statistically significant in all of our specifications. This finding suggests that the recent crisis is different from earlier episodes of financial market stress since 1991 and that this difference is reflected, in part, by an increase in cross-market correlations.

Section 2 discusses our contribution to the literature. Section 3 gives evidence on equity–commodity linkages. Section 4 presents our position data and describes the financialization of commodity futures markets. Section 5 presents our regressions and traces changes in equity–commodity return linkages to fundamentals as well as to hedge fund activity, stress and the interaction of the last two factors. Section 6 concludes.

2. Related work

We contribute to several strands of the financial economics literature. As discussed in the introduction, we provide empirical evidence relevant to theoretical arguments that *who* trades can help predict some aspects of asset return patterns and that the predictive power of trader identity is different during periods of financial market stress. Our findings also place the present paper squarely within a fast-growing literature that analyzes whether commodity prices' levels or distributions are related to the financialization of commodity markets in the past decade.

Several recent papers investigate the impact of financial speculation on commodity prices or returns. Using various techniques, Büyükşahin and Harris (2011), Hamilton (2009), Irwin and Sanders (2012), Korniotis (2009) and Kilian and Murphy (2013) conclude that fundamentals, not speculation, were most likely behind the 2004–2008 boom-bust commodity price cycle.

A number of other studies look at the impact of financialization through the lens of risk premia in commodity markets. Hong and Yogo (2012) argue that the growth – rather than the composition – of the commodity futures open interest helps predict commodity returns. Other studies, though, argue that who trades matters in that the risk-bearing capacities of broker–dealers (Etula, 2010), the risk appetites of commodity producers (Acharya et al., 2013), or the influx of commodity index traders play significant roles in the determination of commodity risk premia around the futures rolling period (Brunetti and Reiffen, 2011) or during the oil boom of 2007–2008 (Singleton, 2013; Hamilton and Wu, 2013).

All the above papers focus on price levels, returns or risk premia. We focus instead on commodity returns' correlation with equity returns. The extent to which this statistic is predicted by the activities of specific types of financial players, and the extent to which commodity returns and other asset returns might be increasingly responding to common factors, are clearly important to commodity market participants who use futures to hedge some underlying commercial exposures. Similarly, correlations matter to portfolio managers' risk-return optimizations and diversification decisions.

Thanks to uniquely aggregated data, we show that the composition of the commodity open interest is an important predictor of cross-market co-movements. In line with the theoretical models mentioned in the introduction, we identify the activities of hedge funds and cross-market traders as relevant to cross-market linkages and we show that, during periods of stress in financial markets, the predictive power of hedge fund positions is weaker.

Related to our query are two earlier studies that exploit publicly available data to investigate the possible impact of commodity index trading (CIT) on cross-commodity return correlations in the past decade. One of those studies finds a CIT impact (Tang and Xiong, 2012); the other concludes that there is no causal relationship (Stoll and Whaley, 2010).

Unlike those papers, our main interest is in the co-movements between commodity and equity markets (rather than the linkages between different commodity futures markets). Our paper further differs with respect to the types of financial traders we analyze (not only index traders but also hedge funds and various other kinds of futures traders). Finally, our paper differs in how we measure financial activity in commodity markets.

Absent other publicly available information, those two studies approximate the total CIT activity in commodity futures markets by extrapolating from information published by the CFTC on CIT positions in 12 agricultural markets. Those data are only available starting in 2006 and, besides, may overstate total CIT activity (see Irwin and Sanders, 2012). In contrast, we utilize the CFTC's non-public trader-level position data for all U.S. markets dating from 2000. These data allow us to identify the daily and weekly shares of commodity futures open interest held not only by CITs but also by hedge funds and several other categories of commodity futures traders.[1]

Using these more detailed data, we find little empirical evidence that commodity-index trading has much to do with changes in equity–commodity co-movements. Our econometric analyses instead suggest that (besides market fundamentals and stress) it is mostly *hedge fund* positions that help predict long-term changes in the strength of equity–commodity linkages.

Our finding that information about *who* trades is less helpful in periods of financial market stress provides empirical support for theoretical papers on limits to arbitrage.[2] It also links our paper to earlier work on financial *vs*. fundamental predictors of cross-border market linkages. Part of that economics literature asks if financial shocks are propagated internationally through financial channels such as bank lending (e.g., van Rijckeghem and Weder, 2001) and international mutual funds (e.g., Broner et al., 2006) or if, instead, shocks spill over through real economy linkages such as trade relationships (e.g., Forbes and Chinn, 2004). Our paper presents an interesting counterpoint in commodity markets by showing that, when the TED spread signals elevated levels of financial stress, higher

[1] Our paper belongs to a growing literature that uses the CFTC's disaggregated, non-public position data. Besides papers already cited, prior work includes Hartzmark's (1987, 1991) studies of the performance of individual traders in nine commodity markets from 1977 to 1981. Leuthold et al. (1994) extend Harzmark. Ederington and Lee (2002) analyze heating-oil futures in 1993–1997. Chang et al. (1997) exploit a dataset of six futures markets from 1983 to 1990. Haigh et al. (2007) analyze hedge funds and oil price volatility in 2003–2004. Büyükşahin et al. (2009) document that increased market participation by hedge funds and commodity index traders helped link crude oil futures prices across the maturity structure. Cheng et al. (2012) study commodity futures traders' responses to financial market stress, proxied by the VIX index.

[2] See Gromb and Vayanos (2010) for a review of the theoretical work on the limits to arbitrage and contagion.

Table 1
Commodity weights.

Year	CBOT wheat	Kansas wheat	Corn	Soybeans	Coffee	Sugar	Cocoa	Cotton	Lean hogs	Live cattle	Feeder cattle	Heating oil	Crude	Natural gas	Copper	Gold	Silver
2000	3.5%	1.3%	4.0%	2.0%	1.3%	2.0%	0.4%	2.2%	3.0%	6.1%	0.0%	7.6%	47.4%	9.6%	7.2%	2.0%	0.2%
2001	4.1%	1.4%	4.4%	2.1%	0.9%	2.2%	0.5%	1.7%	3.1%	6.6%	0.0%	6.8%	47.1%	10.0%	6.9%	2.1%	0.2%
2002	4.5%	1.8%	4.7%	2.4%	0.9%	1.8%	0.7%	1.6%	2.4%	4.9%	0.9%	7.2%	48.4%	8.6%	6.6%	2.4%	0.2%
2003	4.0%	1.5%	4.1%	2.5%	0.8%	1.6%	0.5%	1.9%	2.2%	4.2%	1.0%	6.9%	48.4%	11.7%	6.4%	2.1%	0.2%
2004	3.3%	1.4%	3.6%	2.4%	0.7%	1.3%	0.3%	1.4%	2.1%	3.5%	0.8%	7.7%	51.6%	10.7%	7.1%	2.0%	0.2%
2005	2.4%	0.9%	2.3%	1.6%	0.7%	1.3%	0.2%	1.0%	1.8%	2.7%	0.7%	8.6%	55.8%	11.4%	6.6%	1.7%	0.2%
2006	2.6%	1.0%	2.5%	1.4%	0.7%	1.7%	0.2%	0.9%	1.5%	2.3%	0.6%	8.2%	56.5%	8.1%	9.6%	2.0%	0.3%
2007	3.5%	1.2%	3.2%	1.9%	0.7%	1.1%	0.2%	0.9%	1.4%	2.6%	0.6%	5.9%	57.1%	7.4%	10.1%	2.0%	0.3%
2008	3.8%	0.9%	3.6%	2.2%	0.6%	1.1%	0.2%	0.8%	1.1%	2.2%	0.4%	5.2%	61.8%	6.9%	6.8%	2.0%	0.2%
2009	4.6%	1.0%	4.6%	3.2%	0.9%	2.0%	0.4%	1.0%	1.9%	3.3%	0.6%	4.3%	55.9%	5.5%	6.8%	3.4%	0.4%
2010	4.6%	1.0%	4.6%	3.2%	0.9%	2.0%	0.4%	1.0%	1.9%	3.3%	0.6%	4.3%	55.9%	5.5%	6.8%	3.4%	0.4%

Notes: This table shows the average weights of the 17 GSCI commodities (out of 24 commodities in the index) for which we have trader position data. We use these weights to compute the weighted average measures of trader importance ($WMSS_i$ and $WMSA_i$, where $i =$ AS, AD, AM, AP, MMT, NRP, etc.) as well as the weighted average speculative index (*SIS* and *SIA*). Excluded are four GSCI commodities (aluminum, lead, nickel and zinc) that accounted for less than 5% of the GSCI in 2008 and 2009. The GSCI weight of London Metal Exchange (LME) copper is applied to NYMEX copper positions. Finally, the weight assigned to WTI crude oil is the GSCI weight of WTI crude, plus the weights of Brent crude, gasoil and RBOB gasoline.

hedge fund participation is associated *ceteris paribus* with weaker (rather than stronger) equity–commodity return correlations.

3. Commodity–equity co-movements, 1991–2010

We seek to ascertain whether, in addition to market fundamentals, commodity-market participation by certain types of traders (speculators in general and hedge funds or index traders in particular) helps predict the extent to which a smaller "satellite" asset market (commodities) moves together with a "core" asset market (U.S. equities).

Several papers document fluctuations in the extent to which commodities co-move with one another or with financial assets over time – see, e.g., Erb and Harvey (2006), Gorton and Rouwenhorst (2006), Büyükşahin et al. (2010), Chong and Miffre (2010) and Silvennoinen and Thorp (2013). This section complements and extends those studies beyond the immediate post-Lehman period. It provides summary statistics for equity and commodity index returns and plots estimates of the dynamic conditional correlation (DCC, Engle, 2002) between equity and commodity index returns.

3.1. Commodity and equity return data

We use daily and weekly returns on benchmark commodity and stock market indices.[3] We obtain price data from Bloomberg. Our sample runs from January 1991, when Goldman Sachs' (now Standard and Poor's) GSCI Commodity Index was introduced as an investable benchmark, to March 2010.

For commodities, we use the unlevered total return on the GSCI index, i.e., the return on a "fully collateralized commodity futures investment that is rolled forward from the fifth to the ninth business day of each month." The GSCI includes 24 nearby commodity futures contracts. Because it uses weights that reflect each commodity's worldwide production figures, it is heavily tilted toward energy (see Table 1). In robustness checks, we therefore use another widely used unlevered investable benchmark, the Dow-Jones-UBS Commodity Index Total Return (DJ-UBSCITR; DJ-AIGTR untill May 2009). This rolling index covers 19 physical commodities in our sample period and offers a more "diversified benchmark for the commodity futures market." We find similar results for the GSCI and DJ-UBS indices, and therefore we mainly discuss the GSCI.

For equities, we use Standard and Poor's S&P 500 index. This stock index is broad-based, making it a natural choice. Furthermore, the trading activity in the Chicago Mercantile Exchange's S&P 500 e-Mini futures nowadays far exceeds that of other equity-index futures in the United States, making the S&P 500 e-Mini the ideal market in which to test the hypothesis that cross-market traders may contribute to commodity–equity linkages. We find similar DCC patterns using Dow-Jones' Industrial Average (DJIA) index, and therefore we focus our discussion on the S&P 500.[4] For comparison purposes, we also provide figures for the (generally slightly higher) correlations between the GSCI and the MSCI World Equity index (MSCI).

3.2. Descriptive statistics

Table 2 presents descriptive statistics for the weekly rates of return on the S&P 500 equity index (Panel A) and on the S&P GSCI commodity index (Panel B).

From January 1991 through February 2010, the mean weekly total rate of return on the GSCI was 0.0606% (or 3.16% in annualized terms), with a minimum of −14.59% and a maximum of 14.90%. The typical rate of return varied sharply across the sample period: it averaged 0.14% in 1992–1997 (7.45% annualized); 0.045% in 1997–2003 (2.36% annualized); and 0.0290% in 2003–2010 (1.51% annualized).

[3] Precisely, we measure the percentage rate of return on the Ith investable index in period t as $r_t^I = 100\text{Log}(P_t^I/P_{t-1}^I)$, where P_t^I is the value of index I at time t.

[4] Our equity index returns omit dividends and, hence, underestimate actual returns (Shoven and Sialm, 2000). However, insofar as large U.S. corporations either smooth dividend payments over time (Allen and Michaely, 2002) or do not pay dividends, the correlation estimates that are the focus of our paper should be essentially unaffected.

Table 2
Weekly rates of return – summary statistics (%, January 1991–March 2010).

	1991–2010	1992–1997	1997–2003	2003–2010
Panel A: S&P 500 equity index				
Mean	0.124839	0.272260	0.039058	0.049310
Median	0.292318	0.345607	0.366951	0.247137
Maximum	12.37463	4.194317	12.37463	7.818525
Minimum	−15.76649	−4.112432	−12.18282	−15.76649
Std. dev.	2.348732	1.440671	2.943599	2.371901
Skewness	−0.596507	−0.258227	−0.026150	−1.440103
Kurtosis	7.920789	3.371816	4.791780	10.70166
Jarque–Bera	1066.091***	4.42	42.04***	994.47***
Observations	998	262	314	353
Panel B: S&P GSCI commodity index				
Mean	0.060691	0.138182	0.044902	0.028993
Median	0.188237	0.148651	0.023027	0.416007
Maximum	14.90087	5.340624	7.479387	14.90087
Minimum	−14.59139	−9.208887	−14.59139	−13.12567
Std. dev.	3.023849	1.811528	2.876870	3.870732
Skewness	−0.527095	−0.395102	−0.445674	−0.406732
Kurtosis	5.668868	5.426632	4.888678	4.058999
Jarque–Bera	342.40***	71.10***	57.06***	26.23***
Observations	998	262	314	353

Notes: This table provides summary statistics for the unlevered rates of return on the S&P 500 equity index (excluding dividends; Panel A), as well as on the S&P GSCI commodity index (total return; Panel B). In each panel, the first column uses sample moments computed using weekly rates of return (precisely, changes in log prices multiplied by 100) from January 8, 1991 to March 1, 2010. The second, third and fourth columns use, respectively, weekly rates of returns for three successive sub-periods: May 26, 1992–May 27, 1997; May 27, 1997–May 27, 2003; and, May 27, 2003–February 26, 2010. One, two or three stars indicate that normality of the return distribution is rejected at, respectively, the 10%, 5% or 1% level of statistical significance.

From January 1991 through February 2010, the mean weekly rate of return on the S&P 500 was on average higher than the return on a commodity investment: 0.125% (or 6.71% in annualized terms), with a minimum of −15.77% and a maximum of 12.37%. However, as also noted by Büyükşahin et al. (2010), the rank-ordering of the returns on the two asset classes fluctuates dramatically over time: whereas equities massively outperformed commodities in 1992–1997, the reverse happened in 2003–2008. At the very least, these differences show that equities and commodities do not move in lockstep.

Turning to volatility, Table 2 shows that the rate of return on a well-diversified basket of equities (S&P 500) is generally less volatile than that on commodities (GSCI). The standard deviation of the rate of return on commodities was particularly high after 2003.

3.3. *Dynamic conditional correlations*

Our main interest is in the relationship between commodity and equity returns at various points in time. With unconditional techniques such as rolling correlations or exponential smoothing, the sensitivity of the estimated correlations to volatility changes restricts inferences about the true nature of the relationship between variables, and periods of high volatility only magnify concerns of heteroskedasticity biases – see Forbes and Rigobon (2002). Consequently, we use the dynamic conditional correlation (DCC) methodology of Engle (2002) in order to obtain dynamically correct estimates of the intensity of commodity–equity co-movements.

In essence, the DCC model is based on a two-step approach to estimating the time-varying correlation between two series. First, we estimate time-varying variances using a GARCH(p,q) model. For our sample, $p = q = 1$. Second, we estimate a time-varying correlation matrix using the standardized residuals from the first-stage estimation.

Fig. 1A (*1B*) plots, from January 1991 through February 2010, our estimates of the dynamic conditional correlations between the weekly (*daily*) rates of return on two investable commodity indices (GSCI and DJ-UBS) *vs*. the unlevered rate of return on the S&P 500 equity index. As a benchmark, Fig. 1A (*1B*) also provides a plot for the DCC between the weekly (*daily*) rates of return on the S&P 500 and a second U.S. equity index, the Dow Jones DJIA.

Several facts are clear from Fig. 1A and B. First, in the 18 months following the demise of Lehman Brothers in September 2008, equity–commodity correlations rose to levels never seen in the prior two decades. Second, prior to the Lehman collapse, equity–commodity correlations used to fluctuate substantially over time – but the range was smaller. At both weekly and daily frequencies, the equity–commodity DCC range was −0.38 to 0.4, approaching 0.4 in 1998, 2001–2002, mid-2006, and again in Fall of 2008). Third, despite those ample fluctuations, there is no apparent up-trend in equity–commodity correlations *prior* to August 2008.

3.4. Discussion

Our finding that there was no obvious secular increase in commodity–equity correlations until Fall 2008 is in line with the conclusions of Büyükşahin et al. (2010, p. 78) and Tang and Xiong (2012, p. 55) using weekly or daily data. It is also consistent with findings in Chong and Miffre (2010) and Silvennoinen and Thorp (2013) regarding dynamic correlations between the returns on the S&P 500 and on a number of individual commodity futures. Nevertheless, given that the DCC measure is the dependent variable in the econometric analyses of Section 5, we carry out several robustness checks.

Fig. 1A and B jointly show that the measurement frequency (daily *vs*. weekly) and the choice of commodity index (GSCI *vs*. DJ-UBS) are qualitatively immaterial. Fig. 1C and D likewise show that the choice of equity index (world *vs*. U.S.) does not alter this conclusion.

We use a U.S. stock index (the S&P 500) rather than a global stock market index to compute equity–commodity correlations. We do so for two reasons. One, it minimizes the confounding effects of exchange rate fluctuations on the measurement of commodity–equity co-movements. Two, it allows us to match the correlation we seek to explain with the available equity-futures position data. Suppose, though, that the variable of interest *were* the MSCI–GSCI return co-movements: a comparison of Fig. 1C (1D) with Fig. 1A (1B) shows that we would still find no visible up-trend in equity–commodity correlations prior to September 2008.

Fig. 1E and F, which are based on unconditional rolling correlations, caution that not controlling for time-variations in return volatilities *could* lead to incorrect inferences. First, one might conclude from Fig. 1F (based on unconditional one-year rolling correlations) that GSCI–MSCI rolling correlations strengthened as early as 2006, i.e., well before the Lehman crisis – even though we know from Fig. 1D (based on DCC) that such was not the case. Second, one might conclude from a comparison of Fig. 1E and F (both based on one-year rolling correlations) that the choice of equity index (world *vs*. U.S.) matters – even though we know from a comparison of Fig. 1C (1D) with Fig. 1A (1B) that such is not the case.

Having established that correlations fluctuated substantially, but not dramatically, prior to the Great Recession and having identified a structural break in Fall of 2008), we now turn to the task of predicting the fluctuations depicted in Fig. 1A and B. Do market fundamentals alone predict the observed patterns – or are the latter related to the financialization of commodity futures markets? The next two sections seek to answer these questions.

4. The financialization of U.S. commodity futures markets, 2000–2010

In most U.S. commodity futures markets, the open interest was much greater in 2010 than a decade earlier. In this section, we construct a comprehensive dataset of trader positions in 17 markets and provide novel evidence that this growth entailed major changes in the composition of the overall open interest. We document considerable increases in the presence of hedge funds and in the extent to which equity futures traders are also active in commodity futures markets.

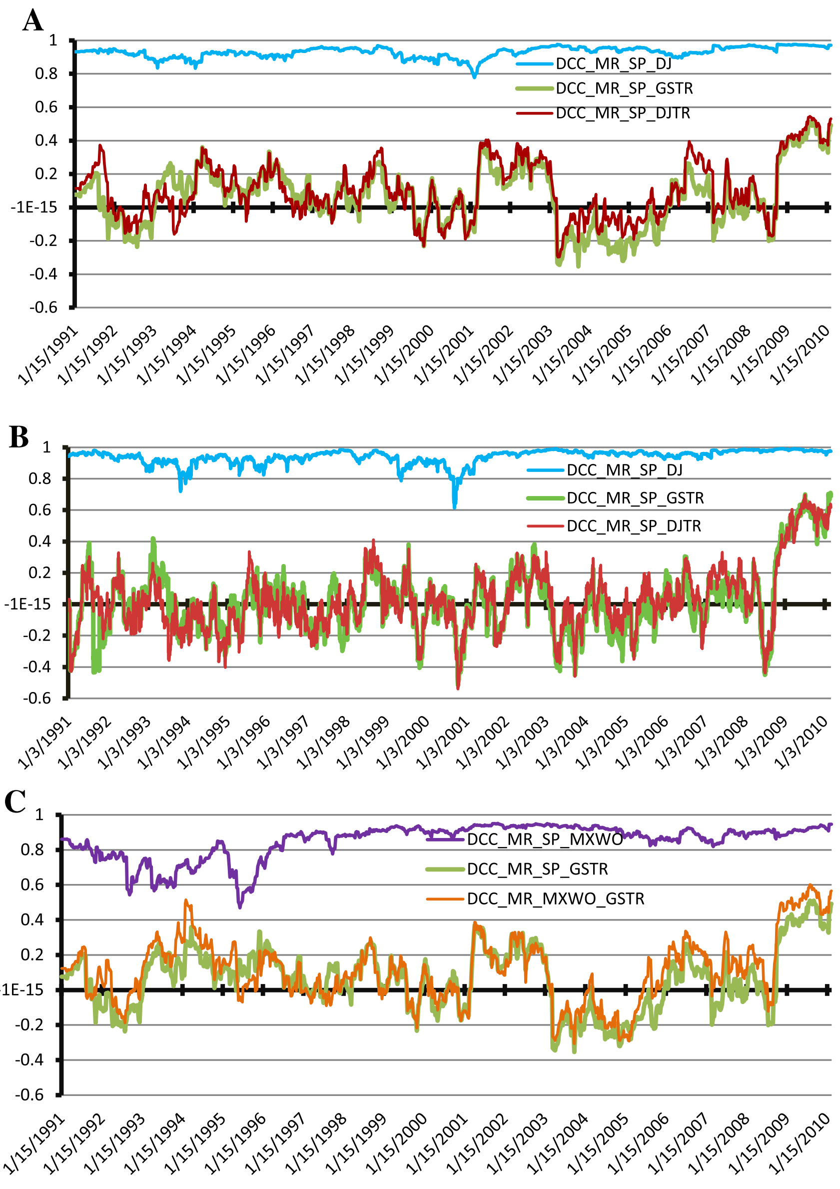

Fig. 1. Return correlations between U.S. equity *vs*. GSCI commodity indices. A: Weekly returns correlations (DCCs), January 1991–March 2010. B: Daily returns correlations (DCCs), January 1991–March 2010. C: Weekly return DCC – world equity *vs*. commodity indices, 1991–2010. D: Daily return DCC – world equity *vs*. commodity indices, 1991–2010. E: Daily return correlations (rolling) – U.S. equity *vs*. commodity indices, 1991–2010. F: Daily return correlations (rolling) – world equities *vs*. commodities, 1991–2010. **Notes:** Figure 1A (*1B*) depicts the time-varying correlation between the **weekly (daily)** unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and: (i) the S&P GSCI total return commodity index (GSTR, **green line**) or (ii) the DJ-UBS total return commodity index (DJTR, **red line**). As a benchmark, the figure also plots the correlation between the S&P 500 equity index and the other traditional equity index, the Dow Jones Industrial Average equity index (DJIA, **blue line** on top). In each case, we estimate dynamic conditional correlation by log-likehood for mean-reverting model (DCC_MR, Engle, 2002) using Tuesday-to-Tuesday returns from January 3, 1991 to March 1, 2010. Figure 1C (*1D*) depicts the dynamic conditional correlations between the **weekly (daily)** unlevered rates of return (precisely, changes in log prices) on S&P's GSCI total return commodity index and: (i) the S&P 500 (SP) equity index (**green line**) or (ii) the MSCI World Equity Index (MXWO, **orange line**). As a benchmark, Figure 1C also plots the correlation between the S&P 500 US equity index and the MSCI World equity index (**purple line** on top). In each case,

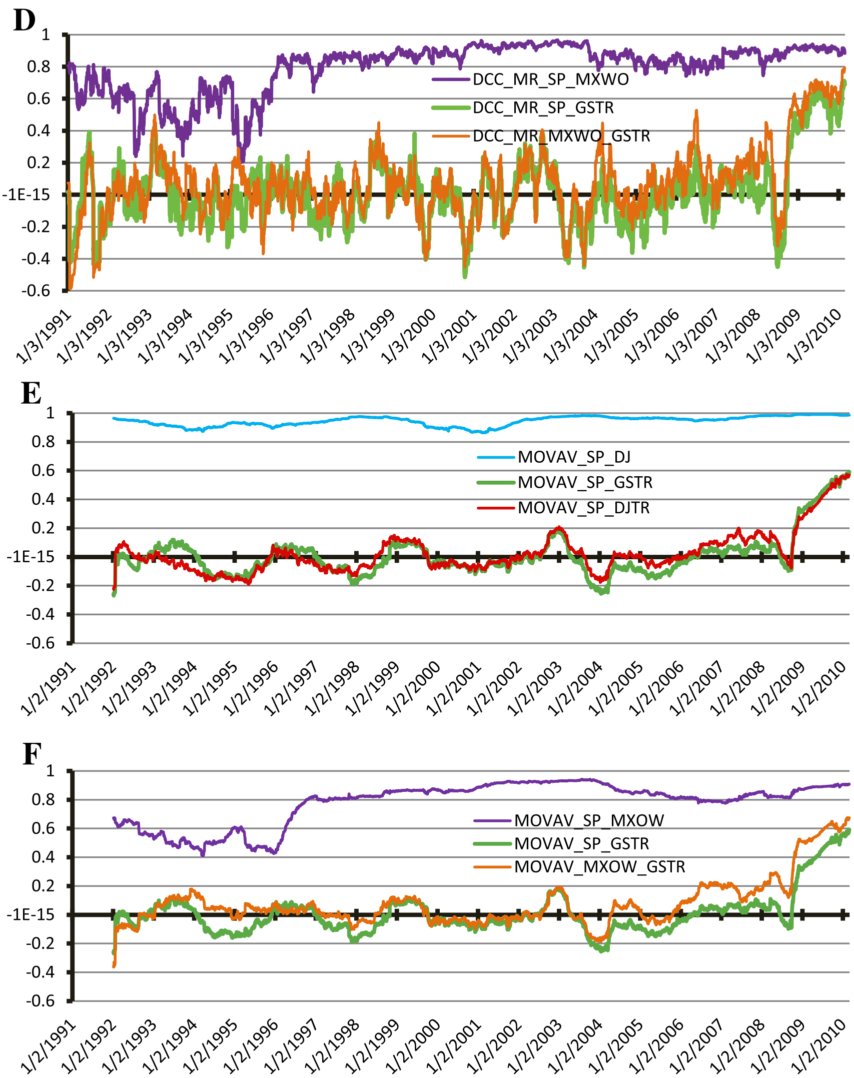

Fig. 1. (*continued*)

dynamic conditional correlation are estimated by log-likehood for mean-reverting model (DCC_MR, Engle, 2002) from January 3, 1991 to March 1, 2010. Patterns are similar, though equity-commodity correlations are slightly greater when estimated with the world equity index rather than with the U.S. equity index. After 2003, the difference between U.S. and world correlations is typically 0.05 to 0.1. Figures 1E and 1F depict **unconditional rolling correlations** between the **daily** rates of return on commodity indices and on U.S. (S&P 500, Fig. 1E) or world (MSCI, Fig. 1F) equity indices. Precisely, Figure 1E depicts one-year rolling correlations between the **daily** unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and: (i) the S&P GSCI total return commodity index (GSTR, **green line**) or (ii) the DJ-UBS total return commodity index (DJTR, **red line**). As a benchmark, the figure also plots the rolling correlation between the S&P 500 equity index and the Dow Jones Industrial Average equity index (DJIA, **blue line** on top). Figure 1F depicts one-year rolling correlations between the **daily** unlevered rates of return (precisely, changes in log prices) on S&P's GSCI total return commodity index and: (i) the S&P 500 (SP) equity index (GSTR, **green line**) or (ii) the MSCI World Equity Index (MXWO, **orange line**). As a benchmark, Figure 1F also plots the correlation between the S&P 500 U.S. equity index and the MSCI World equity index (**purple line** on top). A comparison of, respectively, Figure 1E with Figure 1C and of Figure 1F with Figure 11D shows the importance of controlling for time-varying variances . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

We assemble our dataset by utilizing confidential data on individual trader positions from the U.S. government's futures market regulator: the CFTC. This uniquely detailed information provides the foundation for the regression analyses of Section 5, in which we examine whether participatory changes have explanatory power for equity–commodity returns linkages.

Sections 4.1 and 4.2 describe our dataset and contrast it with the less-detailed (but publicly available) information on futures open interest used in the prior literature.[5] Section 4.3 establishes that, compared to commercial activity, overall speculative activity increased very substantially after 2002. We then disaggregate this information to provide evidence on the growth in these markets of hedge fund activity (Section 4.3), cross-market trading (Section 4.4) and commodity index trading (Section 4.5).

4.1. *Trader position data*

We construct a database of daily trader positions in 17 U.S. commodity futures markets (Table 1 gives a list) and the S&P 500 e-Mini futures market from July 1, 2000 to March 1, 2010.

4.1.1. *Raw data on the purpose and magnitude of individual positions*

The raw position data that we utilize, and the trader classifications on which we rely, both originate in the CFTC's Large Trader Reporting System (LTRS). Specifically, to help fulfill its mission of detecting and deterring market manipulation, the CFTC collects position-level information on the composition of open interest across all futures and options-on-futures contracts for each commodity. It gathers this information for each trader whose position exceeds a certain threshold (which varies by market). It also collects information from each large trader about his respective underlying business (hedge fund, swap trader, commodity producer, etc.) and about the purpose of his positions in different U.S. futures markets.

Many smaller traders' positions are also voluntarily reported to the CFTC and are thus included in the raw data made available for the present study. Our dataset therefore covers more than 75% of the total open interest in each commodity futures market.

The CFTC receives information on individual positions for every trading day. In our weekly analysis, we focus on the Tuesday reports because the underlying raw information is the one which the CFTC summarizes in weekly "Commitment of Traders (COT) Report" that it publishes every Friday at 3:30 pm EST. Consequently, the information we provide in this section can be contrasted with numerous extant studies of commodity markets that rely on COT data.[6]

4.1.2. *Publicly available information*

For every futures market with a certain level of market activity, the CFTC's weekly COT reports provide information on the overall open interest. They also break down this figure between two (until 2009) or four (since 2009) categories of traders.

Prior to September 2009, COT reports separated traders between two broad categories: "commercial" *vs*. "non-commercial." A trading entity generally gets all of its futures and options positions in a given commodity classified as "commercial" by filing a statement with the CFTC that it is commercially "engaged in business activities hedged by the use of the futures or option markets" as defined in CFTC regulations.[7] The "non-commercial" group aggregates various types of mostly financial traders, such as hedge funds, mutual funds, floor brokers, etc.

Since September 4, 2009, COT reports differentiate between four (not two) trader types. They split commercials between "traditional" commercials (producers, processors, commodity wholesalers or

[5] Only a handful of other studies use the CFTC's disaggregated, non-public position data: see footnote 3 in Section 2.

[6] A minor difference is that the large-trader dataset we use includes *all* positions reported to the CFTC – including the positions of numerous traders small enough that they have no regulatory obligation to do so. A second difference is frequency: pre-2000, COTs were published either every two weeks or every month.

[7] In order to ensure that traders are classified accurately and consistently, the CFTC staff may exercise judgment in reclassifying a trader if it has additional information about the trader's use of the markets.

merchants, etc.) and commodity swap dealers (in most markets, this category includes CITs). They also now differentiate between managed money traders (i.e., hedge funds) and "other non-commercial traders" with reportable positions.[8] As of now, however, the CFTC has not indicated plans to make this more detailed information available retroactively prior to 2006 or to break down the aggregate position information by contract maturity.

4.1.3. Non-public information

The LTRS data allow for much more differentiation than the simple COT classifications. Specifically, each reporting trader is classified into one of 28 (rather than a few) sub-categories – commercial dealers, swap dealers, producers, refiners, hedge funds, floor traders, brokers, etc.

Because the LTRS data are contract-specific, they also make it possible to disentangle the activities of various kinds of traders at the near and far ends of the commodity-futures term structure. In contrast, public COT reports do not separate between traders' positions at different contract maturities. Our results in Section 5 show that this additional information is valuable in that it is the positions held by hedge funds in shorter-dated contracts (rather than further along the maturity curve) that contain predictive power for equity–commodity index-return linkages.

An independent contribution of the present paper is thus to provide, in Sections 4.2–4.5, otherwise unavailable information on the composition of open interest in a cross-section of commodity futures markets – in particular, on the positions held by hedge funds since 2000 and on the extent to which equity futures traders have started to trade commodity contracts. We obtained clearance from the CFTC to collect the individual position data for this paper, aggregate it, and present it according to the CFTC's confidentiality statutes.

4.2. Increased "excess speculation"

To gauge the overall growth of speculative activity in U.S. commodity futures markets, we use Working's (1960) "*T*". This index compares the positions of all "non-commercial" commodity futures traders (commonly referred to as "speculators") to the demand for hedging that originates from "commercial" traders (commonly referred to as "hedgers").

4.2.1. Measuring "excess speculation"

Working's "*T*" is predicated on the idea that, if long and short hedgers' respective positions in a given futures market were always exactly balanced, then their positions would always offset one another and speculators might not be needed in that market. In practice, of course, long and short hedgers do not always trade simultaneously or in the same quantity. Hence, speculators must step in to fill the unmet hedging demand. Working's "*T*" measures the extent to which speculation is in "excess" of the level required to offset any unbalanced hedging at the market-clearing price (i.e., to satisfy hedgers' net demand for hedging at that price). Despite its confusingly judgmental name, it is worth noting that "excess" speculation thus defined need not be "excessive" in that at least part of "excess" speculation may be economically necessary in the presence of frictions (the mismatch in timing and maturity of trading contracts).

For each of the 17 commodities in our sample ($i = 1, 2, \ldots, 17$), we calculate Working's T every Tuesday from 2000 to 2010. In each market, we compute two "*T*" indices – one for short-term contracts ($SIS_{i,t}$) only, and one for all contract maturities ($SIA_{i,t}$). The latter measure can be computed using the publicly-available COT reports, which allows readers without access to the LTRS data to replicate this part of our results.

For $SIS_{i,t}$, we use the three shortest-maturity contracts with non-trivial open interest. We do so based on the notion that it is those near-dated contracts whose prices are used to compute the commodity return benchmarks. Formally, in the ith commodity market in week t:

[8] COT reports also provide data on the aggregate positions of non-reporting traders (traders whose positions are too small to require reporting).

$$SIS_{i,t} \equiv T_{i,t} = \begin{cases} 1 + \frac{SS_i}{HL_{i,t}+HS_{i,t}} & \text{if } HS_{i,t} \geq HL_{i,t} \\ 1 + \frac{SL_i}{HL_{i,t}+HS_{i,t}} & \text{if } HL_{i,t} \geq HS_{i,t} \end{cases} \quad (i = 1, \ldots, 17)$$

where $SS_i \geq 0$ is the (absolute) magnitude of the short positions held in the aggregate by all non-commercial traders ("Speculators Short"); $SL_i \geq 0$ is the (absolute) value of all non-commercial long positions; $HS_i \geq 0$ stands for all commercial ("Hedge Short") short positions and $HL_i \geq 0$ stands for all long commercial positions.

We then average these individual index values to provide a general picture of speculative activity across all 17 commodity futures markets in our sample:

$$WSIS_t = \sum_{i=1}^{17} w_{i,t} SIS_{i,t}$$

where the weight $w_{i,t}$ for commodity i in a given *week t* is based on the weight of the commodity in the GSCI index that *year* (Source: Standard and Poor's), rescaled to account for the fact that we focus on the 17 U.S. markets (out of 24 GSCI markets) for which the LTRS position data are available for the entire sample. Table 1 lists the annual commodity weights for each commodity.

To obtain a picture of excess speculation across *all* contract maturities, we also compute:

$$WSIA_t = \sum_{i=1}^{17} w_{i,t} SIA_{i,t}$$

4.2.2. Excess speculation in U.S. commodity futures markets, 2000–2010

Table 3A provides summary statistics of the weighted average speculative indices (*WSIS* and *WSIA*) from July 2000 through February 2010. In that period, the minimum value was 1.11 for both short-term and all contracts; the maximum was 1.5 in near-term contracts (1.42 across all maturities). In other words, speculative positions were on average 11–50% greater than what was minimally necessary to meet net hedging needs at the market-clearing prices.

Fig. 2A documents the growing importance of speculation in commodity markets in the past decade. "Excess" speculation increased substantially, from about 11% in 2000 to about 40–50% in 2008.[9] Interestingly, a comparison of the *WSIS* and *WSIA* curves in Fig. 2A shows that, at almost all times in the sample period, excess speculation was several percentage points greater in near-term contracts than further out on the maturity curve. Notably, excess speculation fell after 2008, especially in near-term contracts (*WSIS* fell from 1.5 to 1.35).

In sum, Fig. 2A identifies a long-term increase, but also substantial variations, in excess commodity speculation. Those patterns will be of particular interest in the analysis of Section 5. Before proceeding to econometric analyses, however, we investigate whether the patterns of overall speculative activity echo the respective patterns for specific categories of financial traders – hedge funds (Section 4.3), index traders (Section 4.4) and cross-market traders (Section 4.5).

4.3. Increased hedge fund activity

Working's *T* lumps together all non-commercial traders: floor brokers and traders, hedge funds, and other non-commercial traders not registered as managed money traders. Yet there is little reason to believe that floor brokers in a specific commodity market should be relevant to equity–commodity linkages. Hedge funds, in contrast, are plausible candidates for such a role.

4.3.1. Measuring hedge fund activity

We exploit the granularity of the LTRS data to compute summary statistics and plot time series of hedge funds' share of the overall commodity futures open interest (see the Appendix for a formal definition of

[9] The values in Fig. 2 are generally lower than historical *T* values for agricultural commodities. Peck (1981) gets values of 1.57–2.17; Leuthold (1983), values of 1.05–2.34. See also Irwin et al. (2008).

Table 3A
Summary statistics on macroeconomic and market fundamentals, July 2000–March 2010.

	Dynamic conditional correlations (DCC_MR)		Macroeconomic fundamentals			Financial market conditions				Excess commodity speculation (working's "*T*")	
	SP500–GSCI	MSCI–GSCI	SHIP index	ADS index	INF (expected inflation)	LIBOR (%)	TED (%)	VIX	UMD	All contract maturities (WSIA)	Short-term contracts (WSIS)
Mean	0.058803	0.124428	0.128101	−0.475016	0.023591	3.058959	0.487749	21.99812	0.003010	1.248605	1.266054
Median	0.051580	0.135190	0.156134	−0.246892	0.023665	2.715900	0.296456	20.41000	0.090000	1.266179	1.264821
Maximum	0.510420	0.602070	0.553002	0.992458	0.033083	6.802500	4.330619	67.64000	4.550000	1.420035	1.499443
Minimum	−0.353440	−0.306940	−0.524973	−3.747359	0.015084	0.248800	0.027512	9.900000	−6.560000	1.112184	1.108470
Std. dev.	0.219529	0.224322	0.263191	0.787462	0.003213	1.874571	0.517985	9.744099	1.127080	0.091597	0.106134
Skewness	0.186990	0.151839	−0.463355	−1.789994	−0.091070	0.328567	2.951072	1.653419	−0.700811	0.097695	0.149171
Kurtosis	1.974010	2.284501	2.329421	6.952640	3.066411	1.842329	14.63722	6.761389	8.153885	1.521860	1.646277
Jarque–Bera	25.09252***	12.71253***	27.53235***	598.4182***	0.790863	37.28642***	3582.557***	527.7928***	600.2571***	46.77718***	40.43314***
Sum	29.69551	62.83637	64.69118	−239.8829	11.91341	1544.774	246.3134	11,109.05	1.520000	630.5457	639.3573
Sum sq. dev.	24.28930	25.36143	34.91194	312.5287	0.005201	1771.064	135.2274	47,853.53	640.2358	4.228546	5.677296
Observations	505	505	505	505	505	505	505	505	505	505	505
ADF (level)	−1.943171	−1.785006	−1.928436	−3.137666**	−1.959157	−1.414196	−2.880949**	−2.995549**	−24.261***	−1.379492	−1.566416
ADF (1st diff)	−22.8378***	−22.9515***	−6.6142***	−12.2230***	−5.7425***	−10.9312***	−12.8887***	−12.3767***	−12.6374***	−22.9845***	−16.8664***

Notes: Dynamic conditional correlations (DCC) are between the Tuesday-to-Tuesday unlevered rates of return (precisely, changes in log prices) on the S&P GSCI total return index (GSTR) and either the S&P 500 equity index (SP) or the MSCI World equity index (MXWO). DCC are estimated by log-likelihood for mean-reverting model (Engle, 2002). *SHIP* is a measure of worldwide economic activity (Kilian, 2009). *ADS* is a measure of U.S. economic activity (Aruoba et al., 2009). *INF* measures expected inflation (source: Federal Reserve). *LIBOR* and *TED* are the 90-day annualized LIBOR rate and TED spread (source: Bloomberg). *UMD* is the Fama-French momentum factor for U.S. equities. Excess commodity speculation for the three nearest-term futures (*WSIS*) and all contract maturities (*WSIA*) is the weighted-average Working "*T*" for the 17 U.S. commodity futures in the GSCI index (source: CFTC, S&P and authors' calculations); annual weights equal the average of the daily GSCI weights that year (source: Standard & Poor). For the augmented Dickey–Fuller (ADF) tests, stars (*, **, ***) indicate the rejection of non-stationarity at standard levels of statistical significance (10%, 5% and 1%, respectively); critical values are from McKinnon (1991). The momentum series is $I(0)$; the others are $I(1)$; the optimal lag length K is based on the Akaike Information Criterion (AIC). Sample period for all statistics: June 26, 2000–February 26, 2010.

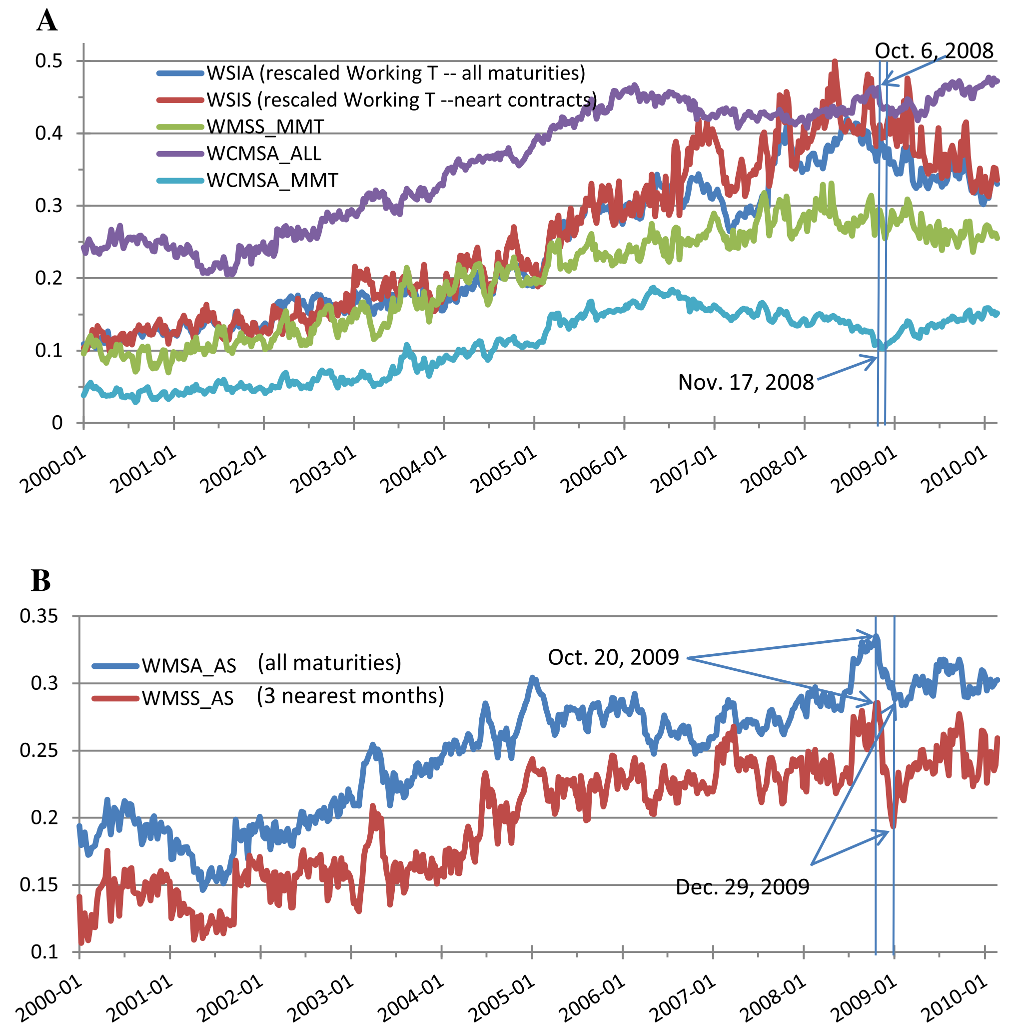

Fig. 2. Financialization of commodity markets. A: Excess speculation, hedge-fund activity and cross-market trading. B: Swap dealing (including commodity index trading). **Notes:** Figure 2A plots the weighted-average speculative pressure index ("Working's T") in the 17 U.S. commodity futures markets linked to the GSCI index in near term contracts (**dark red**, *WSIS*) or across all maturities (**dark blue**, *WSIA*) between the first week of January 2000 and the first week of March 2010. The **green line** shows the aggregate share of the short-term open interest held by hedge funds in the same commodity markets (*WMSS_MMT*). The **purple line** shows the share of the commodity futures positions held by large commodity futures traders that also trade U.S. equity futures (*WCMSA_ALL*), and the **bright blue** line shows the market share of hedge funds that also trade US equity futures (*WCMSA_MMT*). During the same time period, Figure 2B shows the market shares of commodity swap dealers in the same 17 commodity futures markets during the last decade across all contract maturities (**dark blue**, *WMSA_AS*) or in the nearest three months (**dark red**, *WMSS_AS*).

"hedge fund" in U.S. futures markets). We also compute the open-interest shares of commodity swap dealers (a category that includes commodity index traders in most U.S. futures markets – see Section 4.5) and of "traditional" commercial traders. For each sub-category of traders, we compute market shares across the three nearest-maturity futures with non-trivial open interest as well as across all contract maturities.

Formally we compute the "market share" of a given type of traders, in each commodity futures market each Tuesday, by expressing the average of the long and short positions of all traders from this group in that market as a fraction of the total open interest in that market that same Tuesday. We then average these commodity-specific market shares across our 17 commodity futures markets using the commodity weights from Table 1.

We denote by *WMSS_MMT*, *WMSS_AS*, and *WMSS_TCOM* the respective weighted-average market shares of hedge funds (or MMT, "managed money traders"), commodity swap dealers (AS, including CITs – commodity index traders) and traditional commercial traders (TCOM) in short-term contracts. We denote each type of trader's contribution to the total open interest (i.e., across all contract maturities) as *WMSA_MMT*, *WMSA_AS* and *WMSA_TCOM*.

4.3.2. Hedge funds in U.S. commodity futures markets, 2000–2010

The green line in Fig. 2A depicts changes in the *WMSS_MMT* measure over time. This chart, together with Tables 3B and C, highlights several important market changes.

First and foremost, hedge funds' contribution to the commodity futures open interest more than tripled between 2000 and 2008. Their share grew from less than a tenth (*a twentieth*) of the near-term (*overall*) open interest in early 2002 to over a third (*almost 30%*) in early 2008.

Second, Tables 3B and C, which provide summary statistics for various kinds of traders in near-term (Table 3B) and all (Table 3C) futures contracts, show that the market share of floor brokers and traders did not change drastically during the same period. In contrast, Tables 3B and C show that *WMSS_TCOM* and *WMSA_TCOM* ranged from 53% to less than 20%. The implication is that hedge funds' increased market share must echo a sharp drop in traditional commercial traders' proportional contribution to the overall open interest. This conclusion generalizes, to a cross-section of commodity futures markets, some of the observations of Büyükşahin et al. (2009) in the specific case of WTI crude oil futures.

Third, Fig. 2A shows that the market share of hedge funds *as a whole* started trending downward in the second half of 2008. Interestingly, this trend persisted in 2009 and 2010, i.e., in the period when cross-market correlations were unusually elevated.

A natural question is whether *all* hedge funds pulled back from commodities in the post-Lehman turmoil. We debunk this notion in Section 4.4 by showing that one group of hedge funds – those that trade in both commodity and equity markets – in fact increased its collective percentage contribution to the commodity open interest during that period.

4.4. Increased cross-market trading

Of particular interest for this study are commodity futures traders that are also active in equity markets. Table 3D provides information on the number of such traders in each of the commodity futures market in our sample. Fig. 2A and Table 3C document their growing contribution to the overall commodity-futures open interest in the past decade.

4.4.1. Measuring cross-market trading activity

Every reporting trader is uniquely identified in the CFTC's LTRS. For each trading day, we use the unique ID of each commodity futures trader holding open positions at the market close that day to ascertain whether that trader also held overnight positions in the CME's e-Mini S&P 500 equity futures at any point in our sample period. In the affirmative, we consider such a commodity-futures trader to be a "cross-market trader".

This exercise, which we summarize in Table 3D and discuss in Section 4.4.2, tells us how many cross-traders there are on a given trading day. Intuition suggests, however, that traders that are active in both commodity and equity markets likely hold larger positions than do other commodity futures traders. We therefore also compute cross-market traders' share of the overall open interest in a given commodity market on each trading day. To do so, we use the approach of Section 4.3: for each group or subgroup of traders, we compute the open interest attributable to that group or sub-group as the average of the long and short positions of the traders in that group in that market on that day as a fraction of the total open interest in that market on that same day.

We denote by $CMSA\_MMT_{i,t}$, $CMSA\_AS_{i,t}$ and $CMSA\_ALL_{i,t}$ the shares of the open interest in the *i*th commodity held respectively by cross-trading hedge funds (*MMT*), swap dealers (*AS*) and all commodity futures traders (*ALL*) ($i = 1, 2, \ldots, 17$). We then use the commodity weights from Table 1 to

Table 3B
Summary statistics on positions by trader types (short-dated commodity futures), July 2000–March 2010.

	Weighted-average market shares in short-term commodity futures (WMSS)					Weighted-average market share of cross-market traders across all maturities (WCMSA)			
	Hedge funds (WMSS_MMT)	All non-commercials (WMSS_NON)	Swap dealers (WMSS_AS)	Non-commercials + swap dealers (WMSS_ANC)	Traditional commercials (WMSS_TCOM)	All traders (WCMSA_ALL)	Hedge funds (WCMSA_MMT)	All non commercials (WCMSA_NON)	Swap dealers (WCMSA_AS)
Mean	0.205009	0.323114	0.200292	0.523406	0.350287	0.366633	0.108932	0.145403	0.189023
Median	0.219606	0.330037	0.214260	0.547531	0.323999	0.409356	0.115865	0.169236	0.193769
Maximum	0.331050	0.462969	0.285468	0.726409	0.535883	0.476945	0.186921	0.233494	0.262532
Minimum	0.069817	0.176206	0.109090	0.309224	0.191607	0.204867	0.028521	0.049580	0.115570
Std. dev.	0.067763	0.081353	0.042617	0.119861	0.094142	0.083519	0.045061	0.058244	0.030872
Skewness	−0.334458	−0.192197	−0.294154	−0.257259	0.398954	−0.495232	−0.230657	−0.275388	−0.308004
Kurtosis	1.784851	1.671668	1.867022	1.646867	1.866572	1.720597	1.608840	1.492605	2.455440
Jarque–Bera	40.48491***	40.23639***	34.29254***	44.09696***	40.42775***	55.08477***	45.20034***	54.19478***	14.22439***
Sum	103.5296	163.1725	101.1477	264.3202	176.8947	185.1498	55.01070	73.42855	95.45680
Sum sq. dev.	2.314283	3.335603	0.915365	7.240772	4.466850	3.515615	1.023347	1.709729	0.480348
Observations	505	505	505	505	505	505	505	505	505
ADF level	−1.730639	−1.624713	−1.543089	−1.268780	−1.430115	−0.778171	−1.521733	−1.099423	−1.193593
ADF first diff	−16.2192***	−16.5962***	−11.5294***	−20.40521***	−18.5002***	−11.9348***	−17.1272***	−17.5837***	−8.5710***

Notes: *WMSS_MMT*, *WMSS_NON*, *WMSS_AS*, *WMSS_ANC* and *WMSS_TCOM* stand, respectively, for the weighted-average shares of the short-term open interest in the three nearest-dated futures with non-trivial open interest for 17 commodity futures markets of: hedge funds (*MMT*, "managed money traders"), non-commercial traders (*NON*, including MMT), commodity swap dealers (AS, including CITs – commodity index traders), non-commercial plus swap dealers (ANC), and traditional commercial traders (*TCOM*) (source: CFTC and authors' computations). The averaging weights are set each year equal to average of the GSCI weights for those 17 commodities that year and rescaled to account for GSCI commodity markets for which no large trader position data are available (Source: S&P). For three trader types (MMT, AS, NON) as well as all large traders (ALL), the *WCMSA* variables measure the proportion of commodity traders who also hold positions in the S&P 500 e-Mini equity futures ("cross-market traders"). For the augmented Dickey–Fuller (ADF) tests, stars (*, **, ***) indicate the rejection of non-stationarity at standard levels of statistical significance (10%, 5% and 1%, respectively); critical values are from McKinnon (1991). The optimal lag length is based on the Akaike Information Criterion. Sample period for all statistics: June 26, 2000–February 26, 2010.

Table 3C
Summary statistics of positions by trader type (all maturities), July 2000–March 2010.

	Weighted-average market shares in all contracts (WMSA)					Weighted-average market share of cross-market traders across all maturities (WCMSA)			
	Hedge funds (WMSA_MMT)	All non-commercials (WMSA_NON)	Swap dealers (WMSA_AS)	Non-commercials + swap dealers (WMSA_ANC)	Traditional commercials (WMSA_TCOM)	All traders (WCMSA_ALL)	Hedge funds (WCMSA_MMT)	All non-commercials (WCMSA_NON)	Swap dealers (WCMSA_AS)
Mean	0.179422	0.305027	0.251360	0.556387	0.338149	0.366633	0.108932	0.145403	0.189023
Median	0.203755	0.318948	0.263481	0.594333	0.305290	0.409356	0.115865	0.169236	0.193769
Maximum	0.299671	0.426953	0.335178	0.743590	0.537561	0.476945	0.186921	0.233494	0.262532
Minimum	0.056014	0.171564	0.146260	0.327628	0.185196	0.204867	0.028521	0.049580	0.115570
Std. dev.	0.074465	0.079310	0.045016	0.120579	0.096908	0.083519	0.045061	0.058244	0.030872
Skewness	−0.132589	−0.145310	−0.522460	−0.315453	0.408567	−0.495232	−0.230657	−0.275388	−0.308004
Kurtosis	1.543973	1.614327	2.205411	1.764964	1.966243	1.720597	1.608840	1.492605	2.455440
Jarque–Bera	46.08829***	42.17908***	36.25959***	40.47064***	36.53593***	55.08477***	45.20034***	54.19478***	14.22439***
Sum	90.60832	154.0387	126.9370	280.9757	170.7651	185.1498	55.01070	73.42855	95.45680
Sum sq. dev.	2.794682	3.170228	1.021327	7.327757	4.733181	3.515615	1.023347	1.709729	0.480348
Observations	505	505	505	505	505	505	505	505	505
ADF level	−1.379692	−1.440425	−1.193593	−1.043211	−1.135093	−0.778171	−1.521733	−1.099423	−1.517613
ADF first diff	−21.94130***	−23.08369***	−8.5710***	−20.19457***	−21.98602***	−11.9348***	−17.1272***	−17.5837***	−10.59537***

Notes: *WMSA_MMT*, *WMSA_NON*, *WMSA_AS*, *WMSA_ANC* and *WMSA_TCOM* stand, respectively, for the weighted-average shares of the overall futures open interest across *all* futures contract maturities in 17 commodity markets of: hedge funds (*MMT*), non-commercial traders (*NON*, including MMT), commodity swap dealers (*AS*, including *CITs*), non-commercial + swap dealers (*ANC*), and traditional commercial traders (*TCOM*) (Source: CFTC and authors' computations). Weights are set each year equal to the average of the GSCI weights for those 17 commodities that year and are rescaled to account for GSCI commodity markets for which no large trader position data are available (Source: S&P). *WCMSA* variables are as in Table 3A, Panel 2. For the Augmented Dickey–Fuller (ADF) tests, stars (*, **, ***) indicate the rejection of non-stationarity at standard levels of statistical significance (10%, 5% and 1%, respectively). Critical values are from McKinnon (1991). The optimal lag length is based on the Akaike Information Criterion (AIC). Sample period for all statistics: June 26, 2000–February 26, 2010.

Table 3D
Equity–commodity cross-trading activity, 2000–2010.

Commodity	Classifications in commodity markets						Equity futures classification	
	All cross-market traders		Commodity swap dealers		Hedge funds		Hedge funds	
	Count	% of all traders	Count	% of all cross-traders	Count	% of all cross-traders	Count	% of all cross-traders
Cocoa	417	10.5%	29	7.0%	218	52.3%	168	40.3%
Coffee	619	15.6%	37	6.0%	298	48.1%	236	38.1%
Copper	679	17.2%	29	4.3%	290	42.7%	227	33.4%
Corn	786	19.9%	27	3.4%	322	41.0%	256	32.6%
Cotton	587	14.8%	38	6.5%	314	53.5%	245	41.7%
Crude oil	1108	28.0%	63	5.7%	363	32.8%	274	24.7%
Feeder cattle	129	3.3%	15	11.6%	66	51.2%	57	44.2%
Gold	1058	26.7%	43	4.1%	366	34.6%	275	26.0%
Heating oil	335	8.5%	26	7.8%	170	50.8%	138	41.2%
Kansas wheat	251	6.3%	21	8.4%	142	56.6%	114	45.4%
Lean hogs	426	10.8%	23	5.4%	229	53.8%	183	43.0%
Live cattle	469	11.9%	24	5.1%	242	51.6%	196	41.8%
Natural gas	743	18.8%	49	6.6%	300	40.4%	235	31.6%
Silver	604	15.3%	35	5.8%	249	41.2%	201	33.3%
Soybeans	742	18.7%	31	4.2%	305	41.1%	247	33.3%
Sugar	453	11.4%	38	8.4%	230	50.8%	178	39.3%
CBOT wheat	704	17.8%	27	3.8%	311	44.2%	246	34.9%
Median		15.0%		5.9%		49.4%		38.7%

Notes: For 17 commodity futures markets, this table provides information on the number and relative importance of the subset of large commodity futures traders who also held, at some point in the sample period (July 1, 2000–February 26, 2010), positions in the S&P 500 e-Mini equity futures contract.

calculate the weighted-average market share of different types of traders (xxx = *MMT*, *AS* or *ALL*) across the 17 commodity futures markets in our sample:

$$WCMSA\_xxx_t = \sum_{i=1}^{17} w_{i,t} CMSA\_xxx_{i,t}$$

4.4.2. Equity–commodity cross-market activity in U.S. futures markets, 2000–2010

Table 3D provides information the number of cross-market traders, and on the make-up of cross-trading activity, in the 17 commodity futures markets in our sample period. In each of these commodity futures markets, hundreds of traders also held positions in the Chicago Mercantile Exchange's e-Mini S&P 500 equity futures market (Column 1). In all but three of the smallest markets (feeder cattle, Kansas wheat and heating oil), at least 10% of all large commodity futures traders also traded equity futures in that period (Column 2).

Using medians (means are similar), we see that cross-market traders account for 15% of all large commodity futures traders active at some point between July 2000 and March 2010 (Column 2). Hedge funds make up almost 50% (Column 6), whereas commodity swap dealers account for less than 6% (Column 4) of the cross-trading contingent. Approximately 38.9% of all cross-market traders are classified as hedge funds in equity futures markets (Column 8).

These median figures obscure two patterns. One, more than a quarter of all crude oil and gold traders also hold equity futures positions. Two, in contrast, only a seventh or less of all large traders in smaller futures markets ("softs", "livestock" and heating oil) are cross-market traders. In smaller markets, more than half of the cross-market traders are hedge funds while hedge funds make up about a third of all cross-traders in larger commodity markets.

A comparison of Table 3D with the last four columns of Table 3C shows that the median weighted average share of the commodity futures open interest held by equity–commodity cross-traders was

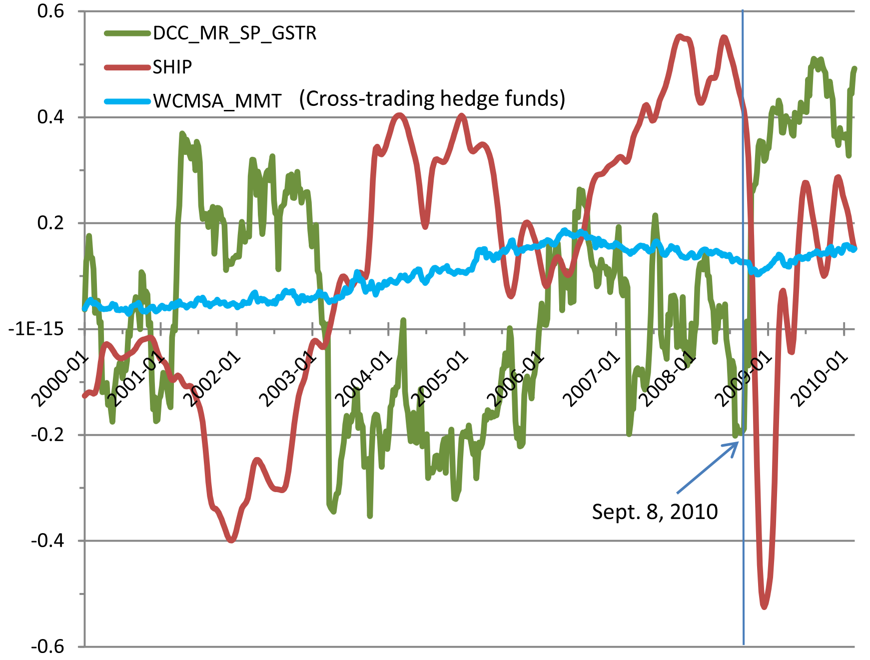

Fig. 3. Equity–commodity correlations, economic activity, and hedge-fund cross-trading. **Notes:** The **green line** in Figure 3 shows, between the first Tuesday of January 2000 and the last Tuesday of February 2010, the dynamic conditional correlation (DCC) between the weekly unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and on the S&P GSCI total return (GSTR) index. We use Tuesday-to-Tuesday return to estimate dynamic conditional correlations by log-likehood for mean-reverting model (*DCC_MR*; Engle, 2002). The dark **red line** (*SHIP*) shows the Kilian (2009) index of worldwide economic activity. The **blue line** shows the weighted-average proportion of hedge funds trading in U.S. commodity futures markets that also trade U.S. equity futures (*WCMSA_MMT*). A negative relationship between *SHIP* and *DCC* is clearly apparent as is, after 2003, a positive long-term relationship between *DCC* and *WCMSA_MMT*.

40.9% during the sample period *vs*. 15% of the trader count. This difference implies that cross-market traders typically hold (much) larger overnight positions than other types of commodity futures traders.

The purple line in Fig. 2A shows that the market share of cross-traders increased substantially between 2000 and 2010, from less than 20% of the total commodity futures open interest in 2000 and 2001 to around 40–47% since mid-2005. The light blue line in Fig. 2A shows that cross-market-trading hedge funds' share of the commodity open interest also grew substantially during that time period.

The magnitude of the latter positions, however, did not move in sync with the positions of other cross-market traders. Most striking is the difference between the activities of hedge funds that trade across markets *vs*. hedge funds that only hold positions in commodity futures markets. As a whole, the market share of hedge funds started a downward trend several months before the Lehman crisis. Notwithstanding some fluctuations, this trend accelerated the week following Lehman's demise. In contrast, cross-trading hedge funds' market share was fairly stable during that period and then *increased* steadily after mid-November 2008.

4.5. Commodity index trading (CIT)

While the non-public data to which we were granted access yields precise information on market shares for most trader categories (including, importantly, for hedge funds), it does not identify CIT activity in energy and metal markets at the daily or weekly frequency. This is because CIT activity percolates into commodity futures markets partly through CIT interactions with commodity swap dealers but, even in the CFTC's non-public LTRS, CIT-related positions cannot be identified within the overall positions held by commodity swap dealers.[10]

One solution to this issue (see, e.g. Stoll and Whaley, 2010; Tang and Xiong, 2012; Singleton, 2013) is to extrapolate, to *all* commodities, CITs' open-interest share in 12 agricultural ("ag") markets – information that has been published by the CFTC, weekly, for those 12 markets since 2006. This approximation, unfortunately, cannot readily be extended to prior years because of structural differences in CIT activity before *vs*. after 2005 (Büyükşahin et al., 2009). Furthermore, after 2006, the quality of that approximation depends on whether the magnitudes of investment flows into commodity markets were similar for ags and for other types of commodities. The precision of the approximation in fact gets worse over time insofar as specialized "ag" funds have grown in importance since 2006 and insofar as the open interest in "ag" futures markets has a different maturity structure than in energy futures markets (Irwin and Sanders, 2012).

We draw instead on the granularity of the non-public CFTC data and on the notion that CIT activity has tended to concentrate in near-dated contracts. Specifically, we proxy the near-term CIT market shares in each of our 17 commodity futures markets each week by the shares of the near-dated open interest held by swap dealers in the same market.[11]

Fig. 2B plots *WMSS_AS* and *WMSA_AS*, i.e., the weighted-average market shares of swap dealers in respectively the three nearest-dated and all commodity futures. For shorter-term contracts in which CIT activity has tended to concentrate (Büyükşahin et al., 2009), Fig. 2B shows that swap dealers' contribution to the commodity open interest increased by about two-thirds between mid-2002 and early-2007. Both *WMSS_AS* and *WMSA_AS* peak in late October 2008 before sharply falling in the following two months. In 2009, both series moved sideways with *WMSS_AS* nearing 25% of the near-dated open interest (a pattern seen from 2007 onward).

5. Economic fundamentals, speculation and commodity–equity co-movements

In Section 3, we showed that the correlation between the weekly returns on investible equity and commodity indices fluctuates substantially over time. In Section 4, we quantified various aspects of U.S. commodity futures markets' financialization in the last decade.

A comparison of Figs. 1A and 2B suggests that the patterns exhibited by swap dealers' positions do not much resemble the equity–commodity returns correlation patterns. Fig. 3, in contrast, suggests that the same is not true for hedge funds positions – especially for the positions of hedge funds that are active in both equity and commodity futures markets.

In this section, we ask formally whether long-term fluctuations in the intensity of speculative activity or in the relative importance of some kinds of trader (in particular, hedge funds) can help predict the extent to which commodity returns move in sync with equity returns. Besides speculative activity, of course, prior literature suggests that economic fundamentals and financial market stress should matter for commodity–equity return correlations. Section 5.1 therefore introduces our real-sector and financial-sector controls. Section 5.2 discusses our ARDL regression methodology, which tackles possible endogeneity issues as well as the fact that some of our variables are stationary in levels while others are only stationary in first differences. Section 5.3 presents our regression results.

Tables 3A and B provide summary statistics for all the variables. Tables 5–7 summarize our regression results.

5.1. Real sector and financial-market conditions

5.1.1. Macroeconomic fundamentals

Business cycle factors affect commodity returns (e.g., Erb and Harvey, 2006; Gorton and Rouwenhorst, 2006). Furthermore, the response of U.S. stock returns to crude oil price increases

[10] Since September 2008, the CFTC has provided quarterly reports about off- and on-exchange commodity index activity in a number of U.S. commodity markets.

[11] An alternative methodology might be to proxy CIT activity by swap dealer positions changes that are common to all near-dated commodity futures.

Table 4
Market fundamentals as long-run predictors of the GSCI-S&P 500 dynamic conditional correlation.

	Model 1			Model 2			Model 3		
	1991–2000	2000–2010	1991–2010	1991–2000	2000–2010	1991–2010	1991–2000	2000–2010	1991–2010
Panel A: treating the post-Lehman period as any other period[a]									
Constant	0.182915** (0.08855)	−0.0676649 (0.1154)	−0.0769808 (0.08335)	0.183340** (0.08916)	−0.0425855 (0.1139)	−0.0456055 (0.07643)	0.983731** (0.3937)	0.198942 (0.5670)	0.266539 (0.2853)
ADS				−0.0192282 (0.09266)	0.136424 (0.1530)	−0.0784245 (0.06634)			
INF							−0.240932** (0.1138)	−0.104337 (0.2242)	−0.118612 (0.09807)
SHIP	−0.03187 (0.2955)	−0.607037** (0.2892)	−0.247640 (0.1946)	−0.0328020 (0.2987)	−0.785661** (0.3811)	−0.249104 (0.1790)	0.268479 (0.2753)	−0.649011** (0.2745)	−0.342061* (0.1947)
UMD	0.0375003 (0.07593)	0.141408 (0.1081)	0.0915841 (0.07970)	0.0399188 (0.07709)	0.126140 (0.1070)	0.0924424 (0.07331)	0.0274178 (0.06095)	0.130264 (0.09883)	0.0872112 (0.07288)
TED	−0.273596 (0.1932)	0.500917** (0.2171)	0.332428** (0.1497)	−0.264308 (0.1967)	0.630212** (0.3125)	0.240228* (0.1410)	−0.268425* (0.1550)	0.476131** (0.2112)	0.288970** (0.1370)
Log likelihood	811.133	855.65	1656.98	811.218	857.236	1658.69	813.068	856.317	1657.7

	Model 1 + DUM		Model 2 + DUM		Model 3 + DUM	
	2000–2010	1991–2010	2000–2010	1991–2010	2000–2010	1991–2010
Panel B: treating the post-Lehman period unlike previous years[b]						
Constant	−0.0314680 (0.06202)	−0.0201306 (0.04778)	−0.00925942 (0.05749)	−0.0193913 (0.04863)	−0.661521 (0.4067)	−0.122517 (0.2137)
ADS			0.153715* (0.08115)	0.00826134 (0.04729)		
INF					0.255100 (0.1599)	0.0363735 (0.07387)
SHIP	−0.407224** (0.1609)	−0.256007** (0.1165)	−0.596757*** (0.1880)	−0.251052** (0.1165)	−0.361970** (0.1539)	−0.229830* (0.1294)
UMD	0.0929004 (0.05706)	0.0695521 (0.04675)	0.0760120 (0.05278)	0.0692592 (0.04678)	0.0762112 (0.05147)	0.0686126(0.04673)
TED	0.201796* (0.1061)	0.110997 (0.08428)	0.334082** (0.1368)	0.111721 (0.08770)	0.211402** (0.09905)	0.113062 (0.08472)
DUM	0.426993*** (0.1220)	0.487423*** (0.1136)	0.485022*** (0.1232)	0.486330*** (0.1243)	0.551450*** (0.1442)	0.518016*** (0.1311)
Log likelihood	859.856	1663.9	862.685	1664.68	861.654	1664.06

[a] Notes to Panel A: The dependent variable is the time-varying conditional correlation (DCC) between the weekly unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and the S&P GSCI total return (GSTR) index. Dynamic conditional correlations are estimated by log-likelihood for mean-reverting model (Engle, 2002). The explanatory variables are described in Table 3A. Long-run estimates are from the two-step ARDL(p,q) estimation approach of Pesaran and Shin (1999). When estimating the long-run relationship, one of the most important issues is the choice of the order of the distributed lag function on y_t and the explanatory variables x_t. The Schwarz information criterion suggests that the optimal lag lengths are $p = 1$ and $q = 1$ in our case. The sample periods for the first, fourth and seventh columns are January 2, 1991–June 30, 2000; for the second, fifth and eight columns: July 1, 2000–February 26, 2010; for the other columns: January 2, 1991–February 26, 2010.

[b] Notes to Panel B: The dependent variable is the time-varying conditional correlation (DCC) between the weekly unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and the S&P GSCI total return (GSTR) index. Dynamic conditional correlations are estimated by log-likelihood for mean-reverting model (Engle, 2002). The explanatory variables are described in Table 3A, except for *DUM* – a time dummy that takes the value $DUM = 0$ prior to September 1, 2008 and $DUM = 1$ afterward ("Lehman dummy"). Long-run estimates are from the two-step ARDL(p,q) estimation approach of Pesaran and Shin (1999). When estimating the long-run relationship, one of the most important issues is the choice of the order of the distributed lag function on y_t and the explanatory variables x_t. The Schwarz Information Criterion (SIC) suggests that the optimal lag lengths are $p = 1$ and $q = 1$ in our case. The sample periods in the first, third and fifth columns are July 1, 2000–February 26, 2010; the sample period for the other columns is January 2, 1991–June 30, 2000.

Table 5A
Speculative activity as a long-run predictors of the GSCI-S&P 500 dynamic conditional correlation.

	2000–2010	2000–2010	2000–2010	2000–2010	2000–2010	2000–2010	2000–2010	2000–2010
Constant	−1.08653***	−3.85024***	−3.42068	−3.10830	−0.582685**	−2.08797**	−3.47333**	−4.18156**
	(0.4039)	(1.328)	(2.801)	(3.180)	(0.2820)	(0.9959)	(1.718)	(1.827)
ADS	0.0900166	0.103863	0.0819206	0.110121	0.121550	0.132858*	0.120917*	0.130751**
	(0.1031)	(0.09492)	(0.1009)	(0.09684)	(0.07598)	(0.07009)	(0.06433)	(0.05701)
SHIP	−1.04023***	−0.933864***	−1.00618***	−0.939143***	−0.716249***	−0.693805***	−0.600611***	−0.496998***
	(0.3210)	(0.2664)	(0.3116)	(0.3016)	(0.2348)	(0.1905)	(0.1989)	(0.1783)
UMD	0.0879904	0.0893486	0.0826899	0.0896561	0.0703609	0.0712289	0.0543494	0.0562307
	(0.07248)	(0.06653)	(0.07076)	(0.06753)	(0.05058)	(0.04622)	(0.04210)	(0.03717)
TED	2.58964**	6.24366**	2.44687**	6.60861*	1.70441**	4.27775**	1.37623**	3.04952*
	(1.076)	(3.120)	(1.041)	(3.448)	(0.6964)	(2.102)	(0.5685)	(1.832)
WMSS_MMT	5.07041***		8.63094**		2.56186*		7.59681***	
	(1.789)		(4.267)		(1.338)		(2.564)	
WMSS_AS			1.84774	−1.68847			1.08052	−1.84586
			(4.015)	(3.330)			(2.453)	(1.914)
WMSS_TCOM			3.54272	−0.879623			4.70689*	1.39127
			(3.967)	(2.595)			(2.487)	(1.502)
WSIA		3.07302***		2.99394		1.64474**		3.24842***
		(1.048)		(1.842)		(0.8008)		(1.057)
INT_TED_MMT	−8.19136**		−7.71792**		−5.22160**	N.A. N.A.	−4.16390**	
	(3.651)		(3.543)		(2.409)		(1.986)	
INT_TED_WSIA		−4.37543*		−4.64281*		−2.96711*		−2.12497
		(2.261)		(2.490)		(1.533)		(1.335)
DUM					0.398321***	0.391434***	0.480248***	0.448907***
					(0.1393)	(0.1273)	(0.1258)	(0.1092)
Log likelihood	867.929	861.365	868.866	862.06	871.537	865.766	874.812	868.424

Notes: The dependent variable (equity–commodity weekly return DCC) and most of the explanatory variables are described in Tables 3A–C. *DUM* is a time dummy for the post-Lehman period (September 2008–March 2010). Long-run estimates are from a two-step ARDL(p,q) estimation (Pesaran and Shin, 1999). The Schwarz Information Criterion (SIC) suggests optimal lag lengths $p = 1$ and $q = 1$. Sample period: July 1, 2000–February 26, 2010.

Table 5B
Cross-market trading as a long-run predictors of the GSCI-S&P 500 dynamic conditional correlation.

	2000–2010	2000–2010	2000–2010	2000–2010	2000–2010	2000–2010
Constant	−0.865047**	−0.506191	−3.77118***	−0.414669	0.755510***	−1.86592***
	(0.3862)	(0.5007)	(1.354)	(0.2550)	(0.2726)	(0.6797)
ADS	0.121594	0.105277	0.103118	0.138361	0.112309**	0.128177***
	(0.1249)	(0.1232)	(0.09608)	(0.08670)	(0.05263)	(0.04763)
SHIP	−0.948281***	−0.898426***	−0.913188***	−0.738715***	−0.484892***	−0.495065***
	(0.3531)	(0.3492)	(0.2732)	(0.2386)	(0.1500)	(0.1365)
UMD	0.0948335	0.0914709	0.0954178	0.0689901	0.0335555	0.0450556
	(0.08516)	(0.08515)	(0.06932)	(0.05648)	(0.03449)	(0.03143)
TED	2.42418**	2.55259**	6.00666*	1.23989*	0.719877*	2.67971*
	(1.173)	(1.209)	(3.148)	(0.7334)	(0.4138)	(1.401)
WCMSA_MMT	7.85698**	9.43273**		3.82713*	4.83782***	
	(3.248)	(4.018)		(2.231)	(1.491)	
WCMSA_AS		−2.94989	−0.256259		−7.06668***	−5.18637***
		(3.406)	(2.633)		(1.756)	(1.459)
WSIA			3.05038**			2.26676***
			(1.210)			(0.5859)
INT_TED_CMMTA	−15.5298*	−16.3698*	N.A. N.A.	−7.24763	−3.52202	
	(8.375)	(8.591)		(5.354)	(3.072)	
INT_TED_WSIA			−4.19862*			−1.85232*
			(2.280)			(1.026)
DUM				0.356019**	0.612165***	0.523899***
				(0.1430)	(0.1104)	(0.09996)
Log lik.	869.216	869.797	862.067	871.446	877.968	870.883

Notes: The dependent variable, most of the explanatory variables and the methodology are described in Table 5A. *INT_TED_CMMTA* and *INT_TED_WSIA* are interaction terms of the TED spread with, respectively, the weekly shares of open interest held by cross-market trading hedge funds (MMT) and swap dealers (AS). Sample period: July 1, 2000–February 26, 2010.

depends on whether the increases are the result of a demand shock or of a supply shock in the crude oil space (Kilian and Park, 2009). These empirical facts point to the need to control for real-sector factors when explaining time variations in the strength of equity–commodity linkages.

To do so, we use a measure of global real economic activity proposed by Kilian (2009), who shows that "increases in freight (shipping) rates may be used as indicators of (…) demand shifts in global industrial commodity markets." The Kilian measure is a global index of single-voyage freight rates for bulk dry cargoes including grain, oilseeds, coal, iron ore, fertilizer and scrap metal. This index accounts for the existence of "different fixed effects for different routes, commodities and ship sizes." It is

Table 6
Long-run predictors of the GSCI-S&P 500 correlations, pre-Lehman.

Variable	Model 6	Model 7	Model 8	Model 9
	2000–2008	2000–2008	2000–2008	2000–2008
Constant	−0.3374 (1.4730)	0.3935 (1.5120)	−0.6700 (2.3950)	2.6124 (2.8380)
SHIP	−0.529*** (0.1688)	−0.5681*** (0.1699)	−0.521*** (0.1709)	−0.6084*** (0.1794)
UMD	0.0254 (0.0363)	0.0253 (0.0366)	0.0219 (0.0369)	0.0150 (0.0376)
TED	0.2094*** (0.0711)	1.198*** (0.4270)	0.203*** (0.0752)	1.4042*** (0.5043)
WMSA_AS	−2.7322 (2.5210)	−4.4055* (2.6040)	−2.6185 (2.6170)	−5.5734* (2.9240)
WMSA_MMT	3.2591* (1.8280)	3.9145** (1.8320)	3.0660 (2.3060)	5.5919** (2.5420)
WMSA_TCOM	0.9390 (1.9110)	−0.4773 (2.0190)	1.0720 (2.0040)	−1.4469 (2.3250)
INT_TED_MMTA		−4.111** (1.6870)		−4.8562** (1.9450)
WSIA			0.2370 (1.4070)	−1.5383 (1.6330)
Observations	437	437	437	437

Notes: Explanatory variables are described in Tables 5A and B. The dependent variable is the time-varying conditional correlation between the weekly unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and the S&P GSCI total return (GSTR) index. Dynamic conditional correlations are estimated by log-likelihood for mean reverting model (Engle, 2002). When estimating the long-run relationship, one of the most important issues is the choice of the order of the distributed lag function on y_t and the explanatory variables x_t. Long-run estimates are from the two-step ARDL(p,q) estimation approach of Pesaran and Shin (1999). The Schwarz Information Criterion suggests that the optimal lag lengths are $p = 1$ and $q = 1$ in our case. The sample period is January 2, 2000–November 11, 2008.

Table 7
Long-run predictors of the GSCI-S&P 500 correlations, pre-Lehman.

Variable	Model 2	Model 3	Model 4	Model 5
	2000–2008	2000–2008	2000–2008	2000–2008
Constant	−2.3482 (1.7890)	−3.2999* (1.7390)	−5.1089** (2.1470)	−5.3486** (2.1140)
SHIP	−0.7083*** (0.1818)	−0.8846*** (0.1846)	−0.6126*** (0.1593)	−0.775*** (0.1738)
UMD	0.0369 (0.0441)	0.0240 (0.0410)	0.0324 (0.0373)	0.0204 (0.0364)
TED	0.3232*** (0.0891)	1.7639*** (0.5417)	0.2329*** (0.0811)	1.4261*** (0.4992)
WMSS_AS	0.8208 (2.6290)	0.7740 (2.4640)	0.9516 (2.2190)	0.9502 (2.1810)
WMSS_MMT	5.3721** (2.6490)	9.143*** (2.9180)	5.0791** (2.2280)	8.2813*** (2.5880)
WMSS_TCOM	2.9873 (2.3960)	3.4884 (2.2690)	4.5071** (2.2210)	4.663** (2.1890)
INT_TED_MMT		−5.924*** (2.1200)		−4.8259** (1.9220)
WSIA			1.803* (0.9239)	1.4199 (0.9360)
Observations	437	437	437	437

Notes: Explanatory variables are described in Tables 5A and B. The dependent variable is the time-varying conditional correlation between the weekly unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and the S&P GSCI total return (GSTR) index. Dynamic conditional correlations are estimated by log-likelihood for mean-model (Engle, 2002). When estimating the long-run relationship, one of the most important issues is the choice of the order of the distributed lag function on y_t and the explanatory variables x_t. Long-run estimates are from the two-step ARDL(p,q) estimation approach of Pesaran and Shin (1999). The Schwarz Information Criterion (SIC) suggests that the optimal lag lengths are $p = 1$ and $q = 1$ in our case. The sample period is January 2, 2000–November 11, 2008.

deflated with the U.S. consumer price index (CPI) and linearly detrended to remove the impact of the "secular decrease in the cost of shipping dry cargo over the last forty years." This indicator is available monthly from 1968. We derive weekly estimates (which we denote *SHIP*) by cubic spline.

Fig. 3, which charts its value from 2000 to 2010, clearly shows an inverse long-term relationship between *SHIP* and our DCC estimates – suggesting that correlations increase when world demand for commodities is low.

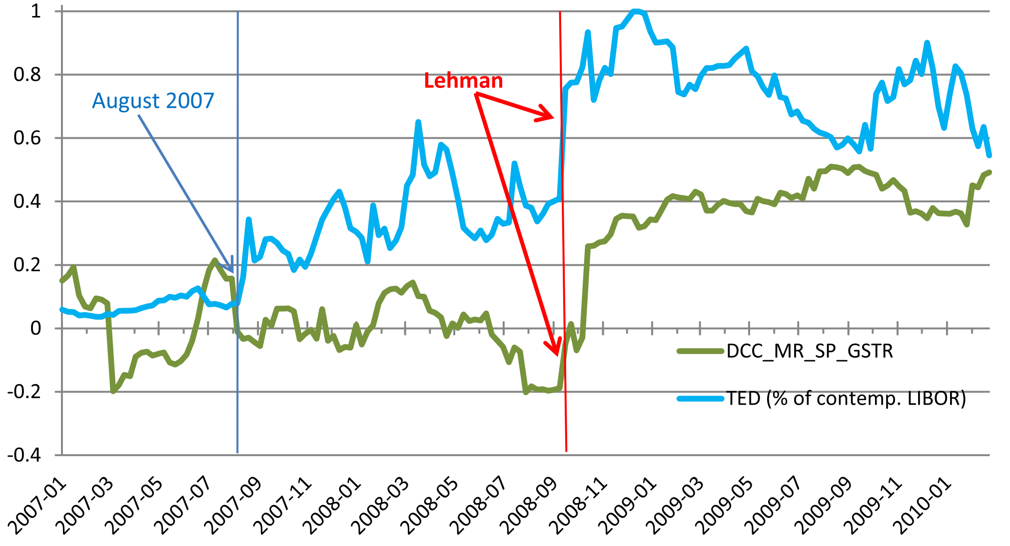

Fig. 4. TED spread and equity–commodity correlations, August 2007–March 2010. **Notes:** The **green line** in Figure 4 shows, between the first Tuesday of July 2007 and the last Tuesday of February 2010, the dynamic conditional correlation between the weekly unlevered rates of return (precisely, changes in log prices) on the S&P 500 (SP) equity index and on the S&P GSCI total return (GSTR) index. We use Tuesday-to-Tuesday return to estimate dynamic conditional correlations by log-likehood for mean-reverting model (*DCC_MR*; Engle, 2002). The **blue line** shows the the 90-day TED spread, expressed as a percentage of the contemporaneous 90-day LIBOR (Source: Bloomberg). Figure 4 shows that this TED-based measure of financial market went up an order of magnitude in the year leading up to Lehman's demise but that equity-commodity correlations did not increase sharply until right after Lehman's demise. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

While SHIP provides a measure of *worldwide* economic activity, U.S. macroeconomic conditions are still central to the prices of U.S. equities and commodities. Consequently, we also consider two macroeconomic variables that may be relevant when studying commodity–equity relationships. One, which we denote *ADS*, is the Aruoba et al. (2009) gauge of U.S. economic activity. This measure is available at weekly frequency for the entire sample period (1991–2010). The other variable captures U.S. inflationary expectations and the intuition that commodities may provide a better hedge against inflation than equities do. We use the figures released each month by the Federal Reserve Bank of Cleveland and carry out a linear interpolation to derive weekly figures (denoted *INF*).

5.1.2. Financial stress and Lehman crisis

Cross-market co-movements increase amid financial stress. Hartmann et al. (2004) identify cross-asset extreme linkages in the case of bond and equity returns from the G-5 countries. Similarly, Longin and Solnik (2001) find that international equity market correlations increase in bear markets. Büyükşahin et al. (2010) show that commodity and equity markets can also behave like a “market of one” during extreme events. We account for this reality in two ways.

First, we include the TED spread (denoted *TED*) in our regressions as a proxy of financial market stress. Table 3A and Fig. 4 show that the TED spread varied widely in our sample period, with a minimum of 0.027% and a maximum of 4.33%.

Second, Fig. 4 shows that the TED spread, though particularly high after the onset of the Lehman crisis, had already started rising – starting in August 2007, when a French financial group froze two funds exposed to the sub-prime market. In contrast, equity–commodity correlations did not *visibly* increase until *after* the demise of Lehman Brothers in September 2008. They remained exceptionally high through the end of our sample in 2010. This difference suggests that the post-Lehman period is exceptional. Hence, we use a time dummy (denoted *DUM*) to account for specificities of this sub-period that the TED spread might not capture.

5.2. Methodology

Before testing the predictive power of different variables on the DCC between equity and commodity returns, we check the order of integration of each variable using Augmented Dickey Fuller (ADF) tests. Unit root tests for the variables in our estimation equation are summarized at the bottoms of Tables 3A and B. They show that some of the variables are $I(1)$ whereas the rest are $I(0)$.

By construction, correlations are bounded above ($+1$) and below (-1), so the DCC variable should intuitively be stationary. Yet the ADF tests do not reject the non-stationarity of the DCC estimates in our sample period. This result holds at the 1% level of significance for the entire sample period (2000–2010, see Table 3A) and at the 10% level of significance for a sub-sample ending prior to the demise of Lehman Brothers (2000 to September 2008).[12]

In order to find the long-run effects of different variables on commodity–equity return correlations, we use an autoregressive distributed lag (ARDL) model estimated by ordinary least squares. In this model, the DCC is regressed on the lags of itself and current and lagged values of a number of regressors (fundamentals as well as traders’ positions). The lagged values of the dependent variable are included to account for slow adjustment of the correlation between commodities and equities. This approach also allows us to calculate the long-run effect of the regressors on the correlation. If our DCC measure is, in fact, stationary, then the ARDL model, estimated by OLS, should give us consistent parameter estimates. If our DCC variable is non-stationary, as suggested by the ADF test statistics, then both short-run and long-run parameters in the ARDL model can still be consistently estimated by OLS if there is a single cointegrating relationship (Pesaran and Shin, 1999).

[12] Because it is well known that ADF tests have low power with short time spans of data, we also employ another test developed by Kwiatkowski et al. (1992) (KPSS) to further analyze the DCC variable. Unlike the ADF test, the KPSS test has stationarity as the null hypothesis. With the KPSS test, we find that the null of stationarity cannot be rejected at the 5% level of significance but is rejected at the 10% significance level.

Specifically, Pesaran and Shin (1999) show that the ARDL model can be used to test the existence of a long-run relationship between underlying variables and to provide consistent, unbiased estimators of long-run parameters in the presence of $I(0)$ and $I(1)$ regressors. The ARDL estimation procedure reduces the bias in the long-run parameter in finite samples and ensures that it has a normal distribution irrespective of whether the underlying regressors are $I(0)$ or $I(1)$. By choosing appropriate orders of the ARDL(p,q) model, Pesaran and Shin (1999) show that the ARDL model simultaneously corrects for residual correlation and for the problem of endogenous regressors.

We start with the problem of estimation and hypothesis testing in the context of the following ARDL(p,q) model:

$$y_t = \delta w_t + \sum_{i=1}^{p} \gamma_i y_{t-i} + \sum_{i=0}^{q} \alpha_i x_{t-i} + \varepsilon_t \tag{1}$$

where y is a $t \times 1$ vector of the dependent variable, x is a $t \times k$ vector of regressors, and w stands for a $t \times s$ vector of deterministic variables such as an intercept, seasonal dummies, time trends, or exogenous variables with fixed lags.[13] In vector notation, Equation (1) is:

$$\gamma(L) y_t = \delta w_t + \alpha(L) x_t + \varepsilon_t$$

where $\gamma(L)$ is the polynomial lag operator $1 - \gamma_1 L - \gamma_2 L^2 - \ldots \gamma_p L^p$; $\alpha(L)$ is the polynomial lag operator $\alpha_0 + \alpha_1 L + \alpha_2 L^2 + \cdots + \alpha_q L^q$; and L represents the usual lag operator ($L^r x_t = x_{t-r}$). The estimate of the long-run parameters can then be obtained by first estimating the parameters of the ARDL model by OLS and then solving the estimated version of (1) for the cointegrating relationship $y_t = \psi w_t + \theta x_t + \nu_t$ by:

$$\widehat{\theta} = \frac{\widehat{\alpha}_0 + \widehat{\alpha}_1 + \cdots + \widehat{\alpha}_q}{1 - \widehat{\gamma}_1 - \widehat{\gamma}_2 - \cdots - \widehat{\gamma}_p}$$

$$\widehat{\psi} = \frac{\widehat{\delta}}{1 - \widehat{\gamma}_1 - \widehat{\gamma}_2 - \cdots - \widehat{\gamma}_p}$$

where $\widehat{\theta}$ gives us the long-run response of y to a unit change in x and, similarly, $\widehat{\psi}$ represents the long-run response of y to a unit change in the deterministic exogenous variable.

When estimating the long-run relationship, one of the most important issues is the choice of the order of the distributed lag function on y_t and the explanatory variables x_t. We carry out a two-step ARDL estimation approach proposed by Pesaran and Shin (1999). First, the lag orders of p and q must be selected using some information criterion. Based on Monte Carlo experiments, Pesaran and Shin (1999) argue that the Schwarz criterion performs better than other criteria. This criterion suggests optimal lag lengths $p = 1$ and $q = 1$ in our case. Second, we estimate the long-run coefficients and their standard errors using the ARDL(1,1) specification.

5.3. Regression results

Tables 5–7 summarize our regression results. Table 4 establishes the predictive power of economic fundamentals (*SHIP* and, to a lesser extent, *ADS*) and financial stress (*TED*). Table 5 establishes the additional predictive power of speculation and hedge fund activities. Table 7 presents some of our robustness checks.

5.3.1. Real sector and financial stress variables

Panels A and B in Table 4 show that, for our sample period (2000–2010) as well as for an extended period (1991–2000, starting when the GSCI first became investable but before the start of our detailed position dataset), the commodity–equity DCC measure is statistically significantly negatively related to *SHIP*. Insofar as *SHIP* captures world demand for commodities, this finding confirms the intuition that cross-market correlations increase in globally bad economic times.

[13] The error term is assumed to be serially uncorrelated.

Our two U.S. macroeconomic indicators (*ADS* and *INF*) have less predictive power. The coefficient for *ADS* is consistently positive but is not always statistically significant. Intuitively, if equities and commodities respond differently to high inflation, then *DCC* and *INF* should be negatively related. Column 7 of Panel A (using data from 1991 to 2010) supports this prediction. In most of our other regressions, however, *INF* is not statistically significant. As Gorton and Rouwenhorst (2006) note, asset returns are volatile relative to inflation; consequently, longer-term correlations better capture the inflation properties of commodity and equity investments. The lack of significance of *INF*, especially in regressions using data from 2000 to 2010 only, may therefore be a mere artifact of sample length.

All of our models include a variable capturing momentum in equity markets (denoted *UMD*). This variable always has a positive coefficient (consistent with the notion that equity momentum could spill over into other risky assets such as commodities), but we never find *UMD* to be a statistically significant predictor of commodity–equity correlations.

The difference between Panels A and B in Table 4 is that the specifications in Panel B include a dummy for the post-Lehman period (*DUM*). That time dummy is always strongly statistically significant and positive, supporting the graphical evidence in Section 3 that this sub-period is exceptional.

Our ARDL estimations show that commodity–equity return correlations also have a positive long-term relationship to the *TED* variable (our proxy for stress in financial markets). In 2000–2010, a 1% increase in the TED spread brought about a 0.20–0.30% increase in the dynamic equity–commodity correlation; this increase is statistically significant at the 5% level of confidence (at the 1% level in 2000–2008; see Table 6).

Interestingly, Panel A suggests that *TED* was not a significant factor in 1991–2000. The difference in the TED spread's importance in those two successive decades suggest a possible predictive role for trading activity. We next turn to this issue.

5.3.2. *Speculative activity and hedge fund market share*

Table 5A is key to our contribution. It shows that trading activity in commodity futures markets helps predict long-term changes in commodity–equity linkages.

Intuitively, there is no reason to expect that traditional commercial traders (oil refiners, grain elevators, etc.) should matter for co-movements between commodity and equity indices. Table 5A confirms this intuition, showing little or no predictive power for *WMSS_TCOM*.

Likewise, insofar as commodity swap dealing overwhelmingly reflects swap dealers' over-the-counter relationships with traditional commercials or with *unlevered*, *long-only*, *passive* commodity-index traders (CITs), we would not expect swap dealers' positions to explain cross-market correlations. This is because CITs do not engage in value-arbitraging and may not alter their positions under financial market stress. Table 5A buttresses this intuition: swap dealers' share of commodity open interest (*WMSS_AS*) is never statistically significantly positive. These findings present an interesting counterpoint to the conclusions of Stoll and Whaley (2010) and Tang and Xiong (2012), both based on public data, regarding intra-commodity market linkages.

The main finding in Table 5A is that, after controlling for economic fundamentals, it is speculative activity in commodity futures markets that helps predict the fluctuations in the commodity–equity DCC estimates over time. *Ceteris paribus*, an increase of 1% in the overall commodity-futures market share of hedge funds (*WMSS_MMT*) is associated with dynamic conditional equity–commodity return correlations that are approximately 4–7% higher (given a mean hedge fund market share of about 25%).

Crucially, Working's "*T*" index of excess speculation in commodity futures markets, which aggregates the activities of *all* non-hedgers across *all* maturities, has less explanatory power than hedge fund activity in short-dated contracts. Precisely, the *WSIA* variable is often but not always significant and, when it is statistically significant, its level of statistical significance is typically lower than that of *WMSS_MMT*. A comparison of likelihood ratios supports this reading – suggesting that it is the positions of hedge funds specifically, rather than the activities of non-commercial traders in general, that help explain the correlation patterns.[14]

[14] Our *WSIA* measure of overall speculative activity could be computed using publicly available COT data. Hence, the results in columns 2 and 6 of Table 5A are consistent with the notion that the changes in Working's *T* can provide, absent access to the CFTC's proprietary disaggregated LTRS data, a useful proxy to capture changes in hedge fund activity measures during the 2000–2010 sample period.

5.3.3. *Cross-market trading*

Table 5B uses specifications similar to Table 5A but focuses on cross-market traders. Two interesting results emerge. First, as intuition would suggest, the open-interest share of hedge funds that trade in both equity and commodity markets helps predict long-term linkages between equity and commodity returns. Second, the open-interest share of commodity swap dealers that are also active in equity markets is sometimes statistically significant – but always with a *negative* sign. These results suggests that it is value arbitrageurs' willingness to take positions in both equity and commodity markets, rather than the trading activities of more traditional commodity market participants, that are relevant to the connection of satellite and central markets.

5.3.4. *Interaction between hedge funds and financial stress*

Table 5 shows that greater hedge fund participation is associated with enhanced cross-market linkages. Yet if the same arbitrageurs or convergence traders, who bring markets together during normal times, face borrowing constraints or other pressures to liquidate risky positions during periods of financial market stress, then their exit from "satellite markets" after a major shock in a "central" market could lead to a decoupling of the markets that they had helped link in the first place.

Building on this hypothesis, some specifications in Table 5 include an interaction term that captures the behavior of hedge funds in financial stress episodes. This interaction term is almost always statistically significant and is always (as expected) negative. That is, *ceteris paribus*, hedge funds' commodity-futures positions are *less* helpful in predicting commodity–equity return co-movements during periods of elevated market stress.

5.3.5. *Implications for portfolio management*

Our results suggest that non-public information on the composition of commodity futures open interest (or, more generally, the make-up of trading activity in financial markets) could be relevant to asset allocation decisions. A corollary is that portfolio managers could benefit from the 2009 CFTC decision to disaggregate the position information that it makes available to the public by separating aggregate trader positions according to traders' principal businesses – hedge fund, commodity-swap dealer, one of several "traditional" commercial types (commodity producer; manufacturer or refiner; wholesaler, dealer or merchant; other), etc.[15]

5.4. *Robustness*

Our results are qualitatively robust to using additional proxies for commodity investment; to introducing dummies to control for unusual circumstances in financial markets; and, to using alternative measures of hedge fund activity in commodity futures markets.

5.4.1. *Commodity indexing activity*

In the past decade, investors have sought an ever greater exposure to commodity prices. Part of this exposure has been acquired through passive commodity index investing. Some of this investment has, in turn, found its way into futures markets through commodity swap dealers. In our regressions, however, we never find the *WMSS_AS* variable (which measures commodity swap dealers' market share in short-dated contracts) to be statistically significant and positive.

One possible reason is that, although a part of commodity swap dealers' positions in short-dated commodity futures reflects their over-the-counter interactions with index traders, the rest of their futures positions reflect over-the-counter deals with more traditional commercial commodity traders. In other words, the *WMSS_AS* variable is only an imperfect proxy of commodity index trading activity in commodity futures markets.

We therefore also used another proxy for investor interest in commodities: the post-2004 daily trading volume in the SPDR Gold Shares exchange-traded fund (ETF). Although this volume grew

[15] It is worth noting that *WMSA_MMT* and *WSIA* (but not *WMSS_MMT*) can, after 2006, be constructed on the basis of the CFTC's COT reports. In other words, some of the information that we show matters is publicly available.

massively between 2004 and 2010, our *GOLD_VOLUME* variable does not help predict changes in commodity–equity correlations.

Taken together with the lack of significance of the *WMSS_AS* variable, our interpretation is that the activities of *passive* commodity investors do not explain equity–commodity linkages. This result presents an interesting counterpoint to the findings of Büyükşahin et al. (2009), who show that increased commodity index trading activity in the WTI crude oil futures market provided additional liquidity that helped integrate crude oil prices across contract maturities.

5.4.2. *Hedge fund activities in near-dated commodity futures* vs. *across the maturity curve*

Table 6 repeats the analysis of Table 5, except that we measure speculative activity and different traders' market shares using position information across all maturities (rather than just the three nearest-maturity contracts with non-trivial open interest). The statistical significance of all the position variables drops dramatically, except for the variable capturing hedge fund activity (*WMSA_MMT* is sometimes significant at the 5% level). Again, Table 6 shows little statistical evidence that swap dealers or traditional commercial traders help predict the dynamic cross-market correlations.

Taken together, Tables 6 and 7 imply that it is the positions of hedge funds in shorter-dated commodity futures (rather than their activities in commodity markets further along the futures maturity curve) that help predict equity–commodity linkages. This result is intuitive, in that the GSCI index is constructed using short-dated futures contracts and, hence, one expects that it is short-dated positions that are relevant in the context of commodity–equity correlations.

5.4.3. *The Lehman crash*

In the last 30 months of the sample period, the TED spread was very high compared to spreads in most of the previous decade. The TED spread first jumped in August 2007; it reached stratospheric levels in September 2008, following the Lehman debacle.

A natural question is whether our results are affected by unusual TED spread patterns during the latter part of our sample period. The answer is negative: our results are qualitatively robust to the introduction of either one of two dummies (one for the August 2007–August 2009 period or one for the September 2008–March 2010 period) and to the concomitant introduction of interaction terms between the relevant dummy and the TED variable.

Table 7 provides additional evidence of robustness. It repeats the analysis of Table 5 with a sample that ends prior to November 2008 – the month when DCC estimates soared upward of 0.4 for the first time since the inception of the investable GSCI commodity index. The results in Table 7 are qualitatively similar to those in Table 5. The main difference is that the statistical significance of the hedge fund variables is stronger pre-crisis. Combined with the statistical significance of the post-Lehman dummy (*DUM*) in every single specification in Table 5, as well as with the negative sign of the *INT_TED_MMT* interaction term, this finding suggests that hedge fund activity *per se* cannot explain the exceptionally high correlation levels observed since the end of 2008.

6. Conclusion and further work

Over the course of the past two decades, the strength of commodity–equity linkages has fluctuated substantially. The last decade also witnessed growing commodity-market activity by hedge funds, commodity index traders, and other financial traders. These facts provide fertile grounds to analyze whether the make-up of trading activity helps predict the joint distribution of commodity and equity returns.

To ascertain whether *who* trades matters for asset pricing, we use non-public trader-level information from the CFTC. We create a daily dataset of all large trader positions in 17 U.S. commodity and equity futures markets from 2000 to 2010. Using this uniquely detailed dataset, we present novel evidence on the financialization of commodity-futures markets. We then document that, besides macroeconomic fundamentals, variations in the composition of the open interest in commodity futures markets do help predict fluctuations in the extent of commodity–equity co-movements.

We trace this predictive power to the activities of speculators in general and hedge funds in particular – especially hedge funds that are active in *both* equity and commodity futures markets. We find that the positions of other kinds of participants in commodity-futures market (swap dealers and index traders, traditional commercial traders, floor brokers and traders, etc.) do not have much predictive power for cross-market correlation patterns, whether or not they take positions in both equity and commodity markets.

We identify two clear patterns when considering the impact of financial market stress on equity–commodity co-movements. First, both before and after Lehman Brothers' demise, we find commodity–equity correlations to be positively related to the TED spread (our proxy for financial stress). Intuitively, hedge funds could be an important transmission channel of negative equity market shocks into the commodity space. In fact, we find that the predictive power of hedge fund positions in commodity futures markets is *lower* in periods of stress.

Second, commodity–equity correlations soared after the demise of Lehman Brothers in Fall 2008 and remained unusually high through Spring 2010. This last finding suggests a natural venue for further research.

Our paper establishes that macroeconomic fundamentals, hedge fund activity, and the TED spread (a proxy for financial market stress) help predict observed long-run fluctuations in commodity–equity correlations. An interaction term between hedge fund activity and TED spread is also significant. Yet, in addition to those other variables, we find that a time dummy for the crisis period (September 2008–March 2010) is always highly significant.

Further research is needed to explain this last finding. One possible explanation is that, amid a crisis of historical proportion and massive uncertainty, a radical shortening of traders' horizons could have made both equities and commodities much more (*less*) sensitive to short-term (*long-term*) economic developments. Another possibility is that the financialization of commodity markets documented in this paper may have made commodity markets more susceptible to "financial market sentiment" – either directly (for example, if collective decisions by passive investors to exit risky markets when uncertainty rises lead to greater correlations between different risky assets) or indirectly (for example, if the prevalence of gloom among too many traders overwhelmed value arbitrageurs' willingness to take on risky positions). A logical question is whether sentiment (interacted or not with our proxies for value arbitraging and for index trading activity) helps predict the increases of commodity–equity or cross-commodity correlations and of the common component of stock returns during the Great Recession.

Finally, our findings highlight the need for theoretical research to help assess whether the empirical relationship that we document (between commodity-market financialization and cross-correlations) represents a welcome improvement in market efficiency or, instead, is a worrisome development. A unifying theory regarding the ideal level of co-movement between commodities and other assets (and between different commodities) is needed – one that would build on prior work, showing the importance of macroeconomic and commodity-specific fundamentals for equilibrium price levels and volatility (Pirrong, 2011) and commodity risk premia (Hirshleifer, 1988), and on models analyzing the impact of different types of traders on cross-market linkages in good or in bad times (e.g., Danielsson et al., 2011, 2013).

Appendix: Defining hedge funds

"Hedge fund" activity in commodity derivatives markets has been the subject of intense scrutiny. Yet there is no accepted definition of a "hedge fund" in futures markets, and there is nothing in the U.S. statutes governing futures trading that defines a hedge fund.

Still, many hedge fund complexes are either advised or operated by CFTC-registered Commodity Pool Operators (CPOs) or Commodity Trading Advisors (CTAs) and Associated Persons (APs) who may also control customer accounts. Through its LTRS, the CFTC therefore obtains positions of the operators and advisors to hedge funds, even though it is not a requirement that these entities provide the CFTC with the name of the hedge fund (or another trader) that they are representing.[16]

[16] A commodity pool is defined as an investment trust, syndicate or a similar form of enterprise engaged in trading pooled funds in futures and options on futures contracts. A commodity pool is similar to a mutual fund company, except that it invests pooled money in the futures and options markets. Like its securities counterparts, a commodity pool operator (CPO) might invest in financial markets or commodity markets. Unlike mutual funds, however, commodity pools may be either long or short derivative contracts. A CPO's principal objective is to provide smaller investors the opportunity to invest in futures and options markets with greater diversification with professional trade management. The CPO solicits funds from others for investing in futures and options on futures. The commodity-trading advisor (CTA) manages the accounts and is the equivalent of an advisor in the securities world.

Clearly, many large CTAs, CPOs, and APs are considered to be hedge funds and hedge fund operators. Consequently, we conform to the academic literature and common financial parlance by referring to these three types of institutions collectively as "hedge funds." We also include in the "hedge fund" category participants not registered in any of these three categories but known to be managing money.

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